

Gustavo Slow State and Capital IQ Quarterly Cycle in PPI and Core-CPI HSA NKPCs

Quarter-on-Quarter Identification and the Prespecified Oil-Control Extension

Satoshi Ichikawa

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Abstract

This report consolidates the estimation sequence undertaken after replacing overlapping year-over-year inflation with annualized quarter-on-quarter PPI and Core CPI inflation. The low-frequency competition environment is anchored by annual Gustavo effective-firm counts, while quarterly Capital IQ concentration information supplies a stationary cycle. The competition measurement posterior is cut from inflation. We then estimate, in order, a constant direct loading, a free static combination of slope and direct channels, a time-varying direct loading, an unrestricted dynamic combination, and the dynamic HSA cross-equation restriction. The report explains why each equation is estimated, documents the data and state construction, reports coefficients and prior-to-posterior learning, plots the implied time-varying coefficients, compares predictive performance, and uses simulation recovery to distinguish computational convergence from economic identification. PPI and Core CPI are estimated separately; they share only the frozen competition-state posterior, sample dates, priors, and model ladder.

The final extension adds current and one-quarter-lagged annualized QoQ changes in the repository's real WTI/CPI oil-price index to every model without changing the competition state or searching over lag specifications. Contemporaneous oil pass-through is tightly positive for PPI and essentially zero for Core CPI. For PPI, the oil control leaves the direct-loading mean nearly unchanged but narrows its posterior and raises its sign probability to about 0.9. The 95% interval still crosses zero, and the slope, time-variation, and HSA proportionality parameters remain unidentified.

The main result is deliberately qualified. PPI average direct loadings are directionally positive, with posterior sign probabilities near 0.8. Core-CPI average direct loadings instead lean negative, while its time-variation coefficient leans positive. Every 95% interval for these competition effects still spans zero. In both price systems the free slope terms, the HSA proportionality coefficient λ , and the derived HSA slope coefficients are not sign-identified. Recovery remains weak at observed effect sizes, and dynamic models do not deliver a stable predictive improvement. All observed fits converge well. The exercise is therefore a successful cross-price computational and identification diagnostic, not an identified structural HSA estimate.

Status. All estimates in this report are diagnostic and not for structural inference. The dynamic branch was run at explicit user request after the predeclared free-static recovery gate failed. A restricted posterior is not treated as independent identification.

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1 Executive Summary

1.1 What changed with QoQ inflation

The original experiment used a four-quarter inflation transformation. Adjacent observations therefore shared three quarters of price changes and required an explicit temporal-error model. The revised primary outcome is

$$\pi_t^q = 400 (\log P_t - \log P_{t-1}), \quad (1)$$

matched to the genuine one-quarter-ahead SPF GDP-price forecast in annualized log units. The primary residual is IID; a stationary persistent AR(1) residual is a robustness check. This makes the dependent variable and expectation horizon correspond to the one-period NKPC.

1.2 What the sequence establishes

1. The QoQ direct loading is updated away from its zero-mean prior: across the four primary cells, $P(\theta > 0)$ is approximately 0.8. Its 95% intervals still include zero.
2. Adding a free slow-state slope interaction does not remove the positive direction of the direct loading. The slope interaction itself is weak.
3. Recovery at observed standardized effects is only 0.10 for the static slope and 0.333 for the static direct loading under the propagated state. Knowing the state exactly does not materially resolve this weakness.
4. Allowing $\theta_t = \theta_0 + \gamma \bar{n}_t^c$ does not identify γ . At an injected standardized $s_\gamma = 0.10$, strong recovery is 0.067.
5. The unrestricted dynamic coefficients and the HSA-restricted coefficients all include zero. In particular, λ is sign-unidentified.
6. All twelve dynamic model/cell combinations have lower 2010Q1–2013Q4 holdout ELPD than constant theta. PSIS-LOO is descriptive because all comparisons contain influential observations with Pareto $k > 0.7$.
7. Core CPI reverses the average-direct direction: $P(\theta > 0)$ is only 0.29–0.38 in M0. Its γ posterior leans positive, reaching $P(\gamma > 0) = 0.86$ in the primary firm-weighted unemployment-gap M2 cell, but every interval still includes zero.
8. Core-CPI recovery and prediction do not validate that sign pattern structurally. At observed effects, propagated-state suggestive recovery is zero for δ , 0.433 for θ , and 0.367 for γ . No dynamic model improves prediction consistently.
9. The prespecified oil control is strongly related to PPI inflation but nearly orthogonal to the competition states. It leaves the PPI M0 mean near 0.69, narrows its interval from $[-0.845, 2.200]$ to $[-0.391, 1.813]$, and raises $P(\theta > 0)$ from 0.816 to 0.890. Core-CPI coefficients are essentially unchanged.

1.3 Bottom line

The aggregate quarterly sample does not yield a price-index-invariant competition effect. PPI suggests a positive average direct loading; Core CPI leans negative for the same loading and positive for its slow-state interaction. None is reliably sign-identified. The valid carried-forward statement is narrower than

“HSA is identified”: the QoQ specification learns about empirical Capital IQ loadings, but the signs are not stable across price indices and the data do not identify the structural mapping from that coordinate to the theoretical mass of active firms. Oil control improves the PPI residual specification and strengthens its directional direct-channel result, but it does not cross the report’s structural identification gate.

2 Why Estimate the Quarter-on-Quarter Equation?

The common NKPC observation equation is

$$\pi_t^q = a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q + \kappa_t x_t - \theta_t \hat{n}_t + \varepsilon_t. \quad (2)$$

Three considerations motivate this form.

Matching the model's time step. The theoretical NKPC is a one-period relation. Annualized QoQ inflation preserves a quarterly time step, whereas YoY inflation is an average of four quarterly inflation rates. A contemporaneous right-hand side combined with YoY inflation would mix horizons.

Matching expectations. The SPF field DPGDP3 is the survey-quarter- t median forecast of annualized GDP-price growth in $t + 1$. It is converted from the SPF discrete annualization to the log convention,

$$E_t \pi_{t+1}^q = 100 \log(1 + r_t^{SPF}/100), \quad (3)$$

which exactly matches $400\Delta \log P_t$. It is a GDP-deflator expectation used as a proxy for PPI expectations, not a price-index-specific PPI forecast.

Removing mechanical overlap. QoQ inflation removes the deterministic three-quarter overlap between adjacent YoY observations. It does not guarantee IID economic shocks; that is why the persistent AR(1) specification is retained. But it prevents mechanical overlap from being mistaken for inflation persistence or competition-channel precision.

Interpretation of signs. For the negative unemployment gap, $x_t = u_t^* - u_t$, higher x_t means a tighter labor market. A conventional Phillips curve therefore has $\kappa_t > 0$. The direct competition term enters as $-\theta_t \hat{n}_t$; under the maintained HSA interpretation, $\theta_t > 0$ is the theoretical sign. No coefficient sign is imposed in estimation.

3 Data, Sources, Transformations, and Sample

The state system begins in 1974Q4, the Capital IQ measurement begins in 1993Q1, and the common NKPC sample is 1993Q2–2013Q4, $T = 83$. The dynamic holdout is fixed at 2010Q1–2013Q4, leaving 67 quarters for training and 16 for evaluation.

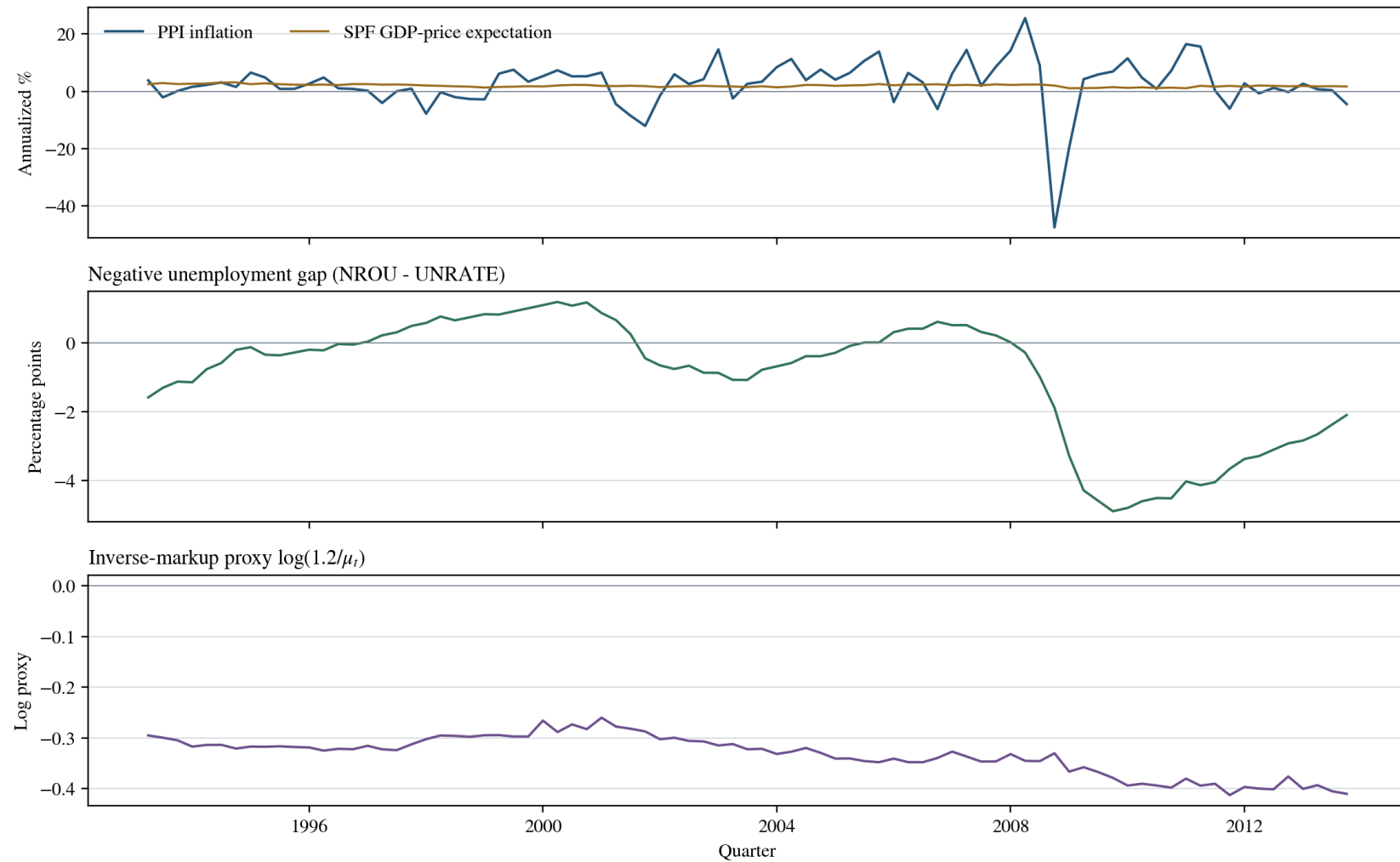
Table 1: QoQ NKPC data summary

Series	Source	Transformation	Start	End	T	Mean	SD
PPI inflation	FRED PPIACO	$400\Delta \log P_t$	1993Q2	2013Q4	83	2.573	8.587
SPF expectation	Philadelphia Fed SPF	$100 \log(1 + r_t/100)$	1993Q2	2013Q4	83	1.988	0.462
Negative unemployment gap	FRED NROU and UNRATE	$u_t^* - u_t$	1993Q2	2013Q4	83	-0.940	1.738
Inverse-markup proxy	Nekarda-Ramey markup	$\log(1.2/\mu_t)$	1993Q2	2013Q4	83	-0.333	0.038

PPI inflation. Monthly FRED PPIACO levels are averaged within quarter and transformed as $400\Delta \log P_t$. The annualized-log convention is applied uniformly in the QoQ design.

Expectations. The Philadelphia Fed SPF workbook `Median_PGDP_Growth.xlsx` provides DPGDP3. The row is dated at survey quarter t and DPGDP3 is the $t + 1$ GDP-price growth forecast. The published annualized percentage rate is converted exactly into annualized log units.

Activity measures. The negative unemployment gap uses quarterly averages of FRED NROU and seasonally adjusted UNRATE. The inverse-markup proxy uses the Nekarda–Ramey business markup μ_t and the calibrated reference $\mu_{ref} = 1.2$: $x_t^{IM} = \log(1.2/\mu_t)$. Its scale is much smaller than the unemployment-gap scale, so raw κ and δ magnitudes cannot be compared across the two activity measures.

QoQ NKPC observables, 1993Q2-2013Q4**Figure 1:** Observed QoQ inflation, expectation, and activity series.

4 Competition Inputs and Their Empirical Meaning

Gustavo annual effective-firm count. The annual series is the inverse of the HHI of U.S. listed firms associated with Grullon, Larkin, and Michaely. It supplies a long-run competition coordinate. The implemented coordinate is

$$g_y = 10 \log(N_y^G / N_{1993}^G). \quad (4)$$

Capital IQ quarterly effective-firm count. Capital IQ firm-quarter revenues for U.S. public firms on major exchanges are aggregated to market HHIs and then to an economy-wide effective number of firms. The report uses both firm-count-weighted and revenue-weighted market aggregation. The coordinate is

$$c_{j,t} = 10 \log(N_{j,t}^{CIQ} / N_{j,1993Q4}^{CIQ}). \quad (5)$$

Why the levels are not equated. The two datasets have different firm universes, industry aggregation, and effective-count scales. Consequently this report does not impose a common observed identity $N_t = \bar{n}_t + \hat{n}_t$ across the two raw series. Instead, Gustavo determines the slow coordinate, while Capital IQ determines a conditional quarterly cycle with an estimated intercept, loading, and measurement error. The notation $\bar{n}, \hat{n}$ describes empirical competition states; it is not yet a proof that the coordinate equals the theoretical HSA mass of active firms.

Competition inputs are related coordinates, not a common observed level

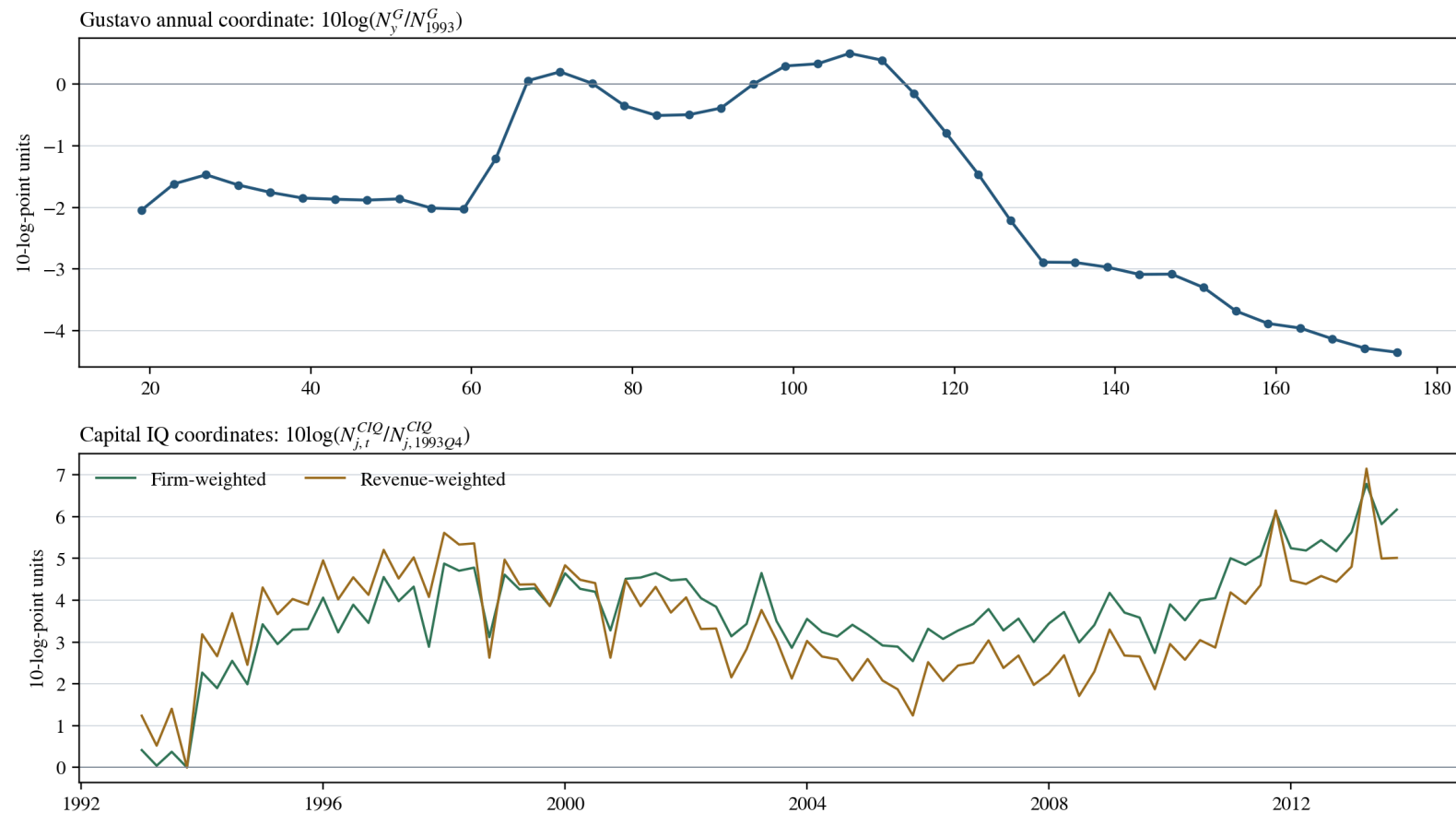


Figure 2: Annual Gustavo and quarterly Capital IQ coordinates.

5 Competition-State Measurement and the Modular Cut

The annual Gustavo observations are exact Q4 endpoints of a quarterly Gaussian bridge,

$$\bar{n}_t = \bar{n}_{t-1} + \mu + \eta_t^{\bar{n}}, \quad (6)$$

$$\bar{n}_{y,Q4} = g_y. \quad (7)$$

Conditional quarterly increments are drawn from the transition law subject to summing to the observed annual change. This supplies a distribution over within-year timing without deterministically interpolating the annual data.

Conditional on a slow-state draw, each Capital IQ coordinate satisfies

$$c_{j,t} = a_j + b_j \bar{n}_t + \hat{n}_{j,t} + e_{j,t}, \quad (8)$$

$$\hat{n}_{j,t} = 2r_j \cos(2\pi/P_j) \hat{n}_{j,t-1} - r_j^2 \hat{n}_{j,t-2} + u_{j,t}. \quad (9)$$

The bounds $r_j \in [0.20, 0.95]$ and $P_j \in [6, 24]$ quarters keep the cycle stationary and exclude an arbitrary second trend.

The ordering is

$$p(\bar{n} \mid G) p(\hat{n}, a, b, r, P, \sigma_h, \sigma_e \mid CIQ, \bar{n}) p(\text{NKPC coefficients} \mid \pi, x, E\pi, \bar{n}, \hat{n}). \quad (10)$$

Inflation cannot update either competition state. This modular cut prevents the state-space model from constructing a cycle merely because it helps explain inflation.

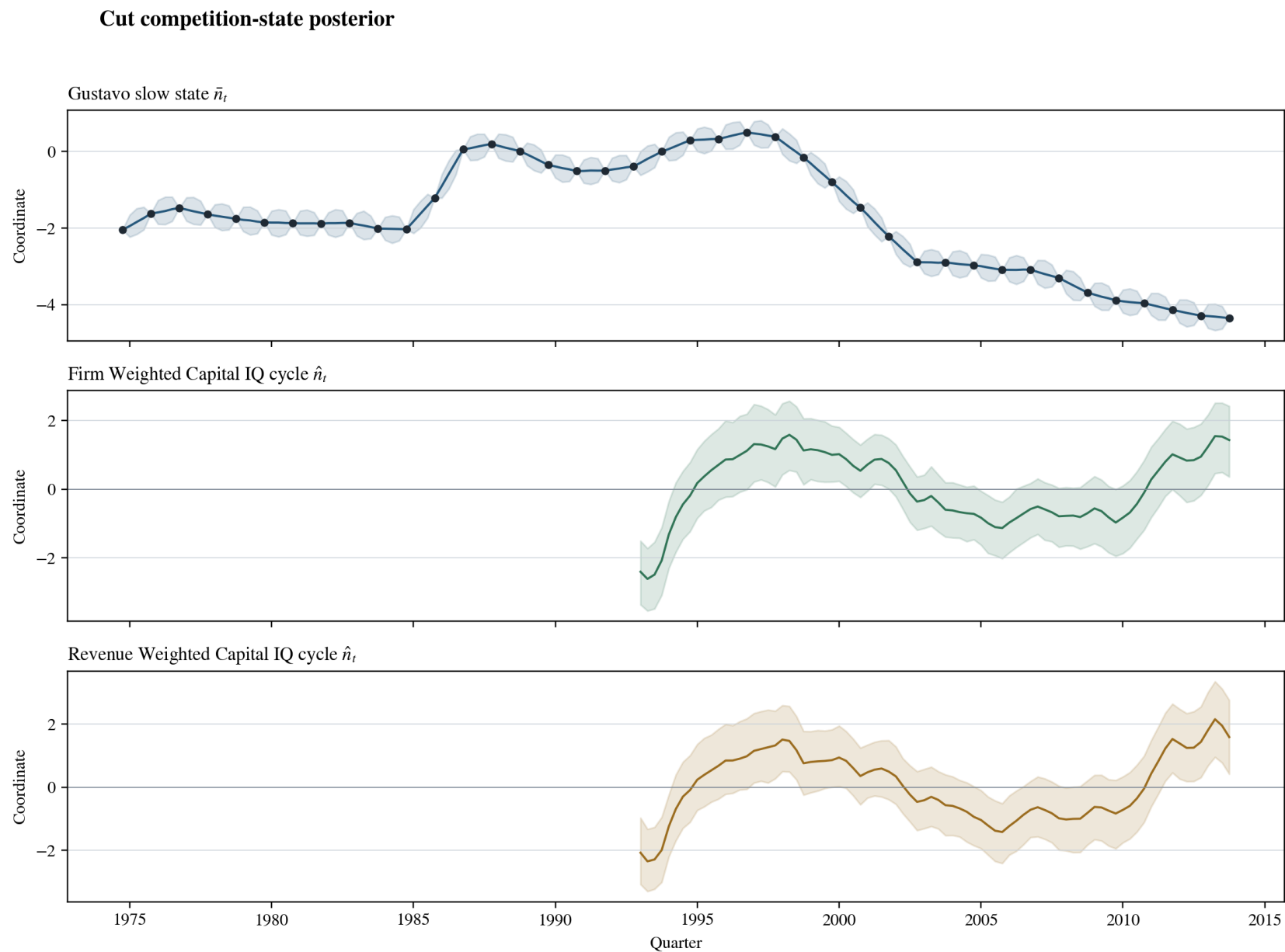


Figure 3: Slow and cyclical competition-state posterior. Bands are pointwise 95% intervals.

6 Why the Models Are Estimated in This Order

The estimation sequence is identification-first rather than restriction-first.

1. **Constant θ / direct only:** ask whether the quarterly Capital IQ cycle has a free direct loading before using any slow-state slope information.
2. **Free static combined:** add a free slow slope δ and check whether θ survives, without linking them.
3. **Recovery before restriction:** inject known δ and θ into the actual 83-quarter design and test whether the unrestricted model can recover them.
4. **Varying θ :** add γ without adding dynamic slope coefficients. This isolates whether the direct loading itself varies.
5. **Free dynamic:** allow linear and quadratic slope variation independently.
6. **HSA-restricted dynamic:** only then impose the cross-equation restrictions and compare their predictive implications.

This ordering matters. If the free coefficients cannot be recovered, a product restriction can create a narrower posterior by transferring information across equations. Such narrowing is an implication of the restriction, not evidence that the direct channel was independently identified.

The predeclared free-static gate failed and would ordinarily stop the dynamic branch. The dynamic models in this report were estimated afterward at explicit user request as a weak- identification diagnostic. They must be interpreted in that status.

7 Estimated Model Hierarchy

Let

$$\bar{n}_t^c = \bar{n}_t - T^{-1} \sum_s \bar{n}_s, \quad q_t^{(2)} = (\bar{n}_t^c)^2 - T^{-1} \sum_s (\bar{n}_s^c)^2. \quad (11)$$

Centering keeps κ_0 and θ_0 at the sample-average slow state. Centering the quadratic term changes only the intercept of $\kappa(\bar{n})$, not its derivative.

7.1 M0: constant theta / direct only

$$\kappa_t = \kappa_0, \quad \theta_t = \theta. \quad (12)$$

This is the minimum free-direct diagnostic.

7.2 M1: free static combined

$$\kappa_t = \kappa_0 + \delta \bar{n}_t^c, \quad \theta_t = \theta. \quad (13)$$

This asks whether the slow slope and cyclical direct channels coexist. It is not HSA- restricted because δ and θ are independent.

7.3 M2: varying theta

$$\kappa_t = \kappa_0, \quad \theta_t = \theta_0 + \gamma \bar{n}_t^c. \quad (14)$$

This isolates time variation in the direct channel.

7.4 M3: free dynamic

$$\kappa_t = \kappa_0 + \delta_1 \bar{n}_t^c + \delta_2 q_t^{(2)}, \quad \theta_t = \theta_0 + \gamma \bar{n}_t^c. \quad (15)$$

It is the unrestricted benchmark for the HSA dynamic restriction.

7.5 M4: HSA-restricted dynamic

HSA links slope and direct channels through

$$\frac{d\kappa(N)}{dN} = \lambda \theta(N), \quad \lambda = b_x \zeta. \quad (16)$$

With $\theta(N) = \theta_0 + \gamma N$, integration gives

$$\kappa_t = \kappa_0 + \lambda \theta_0 \bar{n}_t^c + \frac{\lambda \gamma}{2} q_t^{(2)}, \quad (17)$$

so the tested restrictions are

$$\boxed{\delta_1 = \lambda \theta_0, \quad \delta_2 = \lambda \gamma / 2.} \quad (18)$$

The sign of λ is not fixed. Because Capital IQ is an empirical concentration coordinate and b_x is not externally identified here, λ is a proportionality parameter, not automatically a demand elasticity.

8 Priors, Sampling, and Identification Criteria

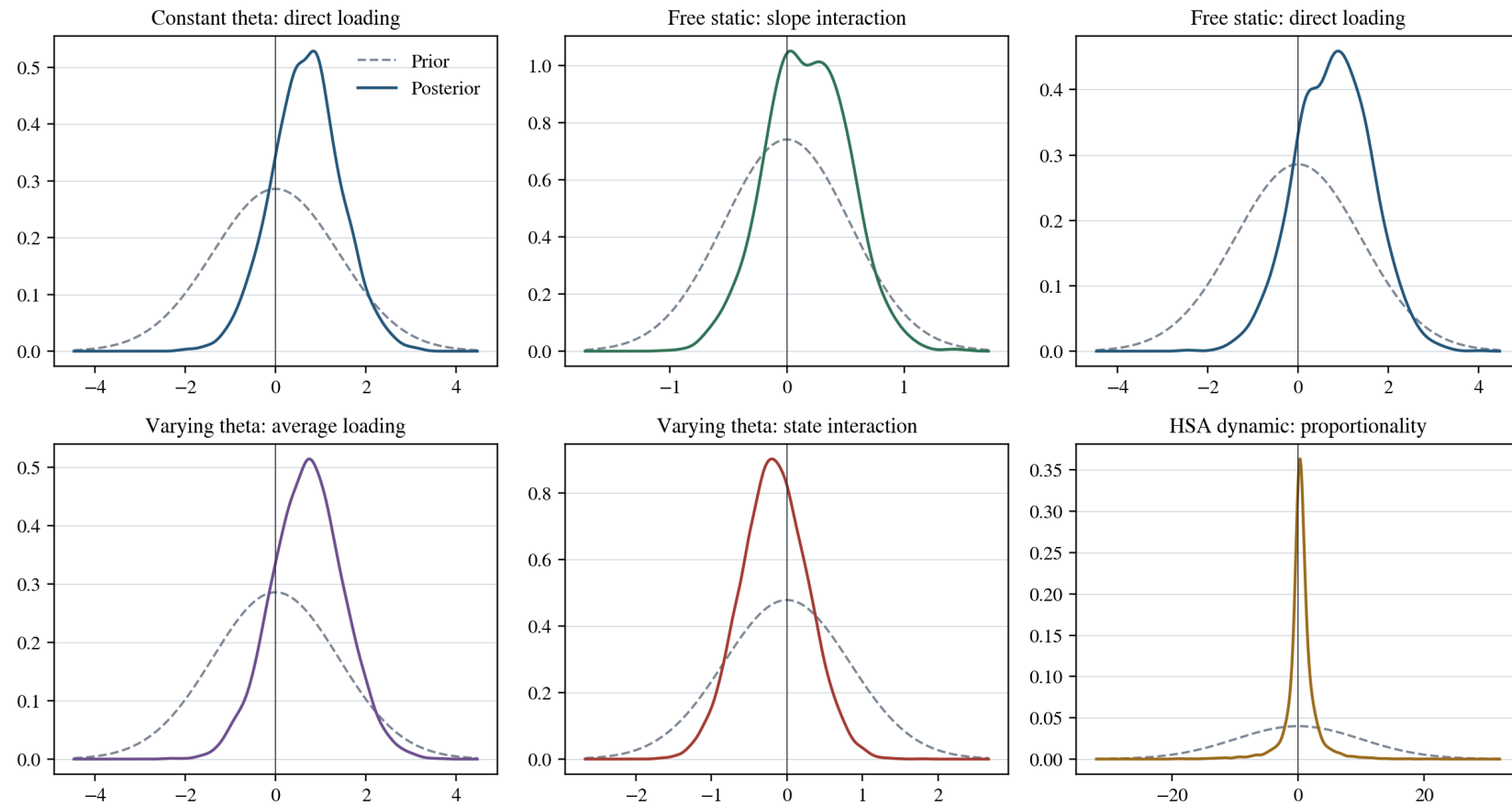
For every coefficient whose regressor is z_t , the competition prior is standardized so one prior standard deviation contributes 0.20 inflation standard deviations:

$$\beta_z \sim N\left(0, \left[0.20 \frac{s_\pi}{s_z}\right]^2\right). \quad (19)$$

The intercept prior is $N(0, (2s_\pi)^2)$, $\alpha_b, \alpha_f \sim N(0.5, 0.5^2)$, and $\lambda \sim N(0, 10^2)$. No economic sign is imposed. The innovation variance has the common inverse-gamma update with shape 3 and scale $2s_\pi^2$. The AR(1) robustness prior is $\rho \sim N(0, 0.5^2)$ truncated to $[-0.95, 0.95]$.

Constant-theta and free-static observed fits use four chains, 2,000 iterations, 600 warmup, and thinning by two. The three dynamic specifications use four chains, 5,000 iterations, 1,500 warmup, and thinning by two. Observed dynamic fits require maximum R-hat at most 1.05 and minimum bulk ESS at least 400.

Learning versus sign evidence. Posterior narrowing is summarized by SD_{post}/SD_{prior} . Suggestive recovery requires a correct-sign probability of at least 0.80 and an SD ratio at most 0.75. Strong recovery also requires sign probability 0.975 and a 95% interval excluding zero. Convergence, learning, sign identification, and predictive performance are reported separately.

Prior versus posterior: primary firm-weighted unemployment-gap cell**Figure 4:** Selected prior-versus-posterior densities.

9 Competition Measurement Results

Table 2: Competition-state parameters; posterior mean and 95% interval

Block	Variant	Parameter	Posterior
gustavo_slow	gustavo	mu	-0.015 [-0.045, 0.015]
gustavo_slow	gustavo	sigma_bar	0.179 [0.146, 0.222]
capital_iq_cycle	firm_weighted	intercept	2.860 [2.038, 3.704]
capital_iq_cycle	firm_weighted	loading	-0.383 [-0.761, -0.061]
capital_iq_cycle	firm_weighted	damping	0.779 [0.617, 0.892]
capital_iq_cycle	firm_weighted	period	20.204 [15.715, 23.735]
capital_iq_cycle	firm_weighted	sigma_cycle	0.310 [0.181, 0.538]
capital_iq_cycle	firm_weighted	sigma_measurement	0.462 [0.352, 0.576]
capital_iq_cycle	revenue_weighted	intercept	3.442 [2.541, 4.122]
capital_iq_cycle	revenue_weighted	loading	0.027 [-0.295, 0.330]
capital_iq_cycle	revenue_weighted	damping	0.756 [0.600, 0.879]
capital_iq_cycle	revenue_weighted	period	19.436 [14.752, 23.643]
capital_iq_cycle	revenue_weighted	sigma_cycle	0.363 [0.212, 0.568]
capital_iq_cycle	revenue_weighted	sigma_measurement	0.660 [0.528, 0.814]

The exact Gustavo Q4 anchor error is below 10^{-15} . The slow innovation scale is 0.179[0.146, 0.222] in the ten-log-point coordinate. Capital IQ cycles are persistent but stationary: damping is about 0.76–0.78 and the posterior periods are about 19–20 quarters.

The most consequential measurement result is the slow loading. Firm weighting gives $b = -0.383[-0.761, -0.061]$, whereas revenue weighting gives 0.027[−0.295, 0.330]. The two datasets therefore do not support a stable one-for-one level mapping. The negative firm loading is not interpreted as a structural negative relation; it may reflect listed-firm coverage, entry/exit composition, market weighting, or industry aggregation differences. This is precisely why the report gives the two datasets distinct measurement roles.

The cycle measurement error is also material: σ_e is 0.462 for firm weighting and 0.660 for revenue weighting. Propagating this uncertainty is empirically important; treating the posterior median cycle as known would overstate precision.

10 M0 Results: Constant Direct Loading

The estimated equation is displayed again next to its results:

$$\pi_t^q = a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q + \kappa_0 x_t - \theta_{CIQ} \hat{n}_t + \varepsilon_t. \quad (20)$$

Table 3: Constant-theta model; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
alpha_b	0.342 [0.139, 0.545]	0.344 [0.139, 0.548]	0.339 [0.131, 0.545]	0.336 [0.134, 0.547]
alpha_f	0.393 [-0.566, 1.328]	0.407 [-0.527, 1.368]	0.412 [-0.540, 1.356]	0.406 [-0.528, 1.326]
kappa_0	-0.019 [-0.943, 0.897]	-0.077 [-0.986, 0.793]	-11.227 [-48.249, 28.316]	-12.844 [-51.463, 25.927]
theta_CIQ	0.674 [-0.845, 2.200]	0.609 [-0.783, 2.055]	0.656 [-0.843, 2.176]	0.669 [-0.762, 2.065]

The direct coefficient is positive on average in all four cells. Posterior positive probabilities are approximately 0.79–0.84, and posterior SD is about 54–57% of prior SD. This is genuine updating from a symmetric zero-mean prior. It is not, however, strong sign identification: every 95% interval includes zero.

The backward inflation coefficient is well learned near 0.34. In contrast, the forward coefficient remains close to its $N(0.5, 0.5^2)$ prior in dispersion. This is another source of uncertainty: the expectation term can absorb low-frequency variation that might otherwise help estimate the competition loadings.

The inverse-markup κ_0 is numerically large because the inverse-markup proxy has a sample SD of only 0.038. It must not be compared directly with the unemployment-gap slope.

11 M1 Results: Free Static Combined Channels

The free-static equation is

$$\pi_t^q = a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q + (\kappa_0 + \delta \bar{n}_t^c) x_t - \theta_{CIQ} \hat{n}_t + \varepsilon_t. \quad (21)$$

Table 4: Free-static combined model; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.198 [-1.055, 1.475]	0.081 [-1.110, 1.290]	-9.952 [-52.861, 36.946]	-11.476 [-56.517, 29.453]
delta	0.173 [-0.517, 0.861]	0.141 [-0.567, 0.874]	0.221 [-2.779, 3.255]	0.214 [-2.813, 3.173]
theta_CIQ	0.784 [-0.832, 2.388]	0.710 [-0.802, 2.198]	0.672 [-0.874, 2.189]	0.630 [-0.851, 2.063]

Adding the slow slope interaction does not eliminate the direct update. In the two primary IID unemployment-gap cells, $P(\theta_{CIQ} > 0)$ is about 0.83. The largest absolute posterior correlation between δ and θ_{CIQ} is 0.352, below the predeclared severe- confounding threshold of 0.80.

But every δ interval includes zero. The fixed 2010Q1–2013Q4 holdout ELPD is lower for free static in all four cells, and the Savage–Dickey BF_{01} lies between 1.38 and 1.57, mildly favoring $\delta = 0$. Thus the slow slope term does not explain away theta, but neither does it earn empirical support of its own.

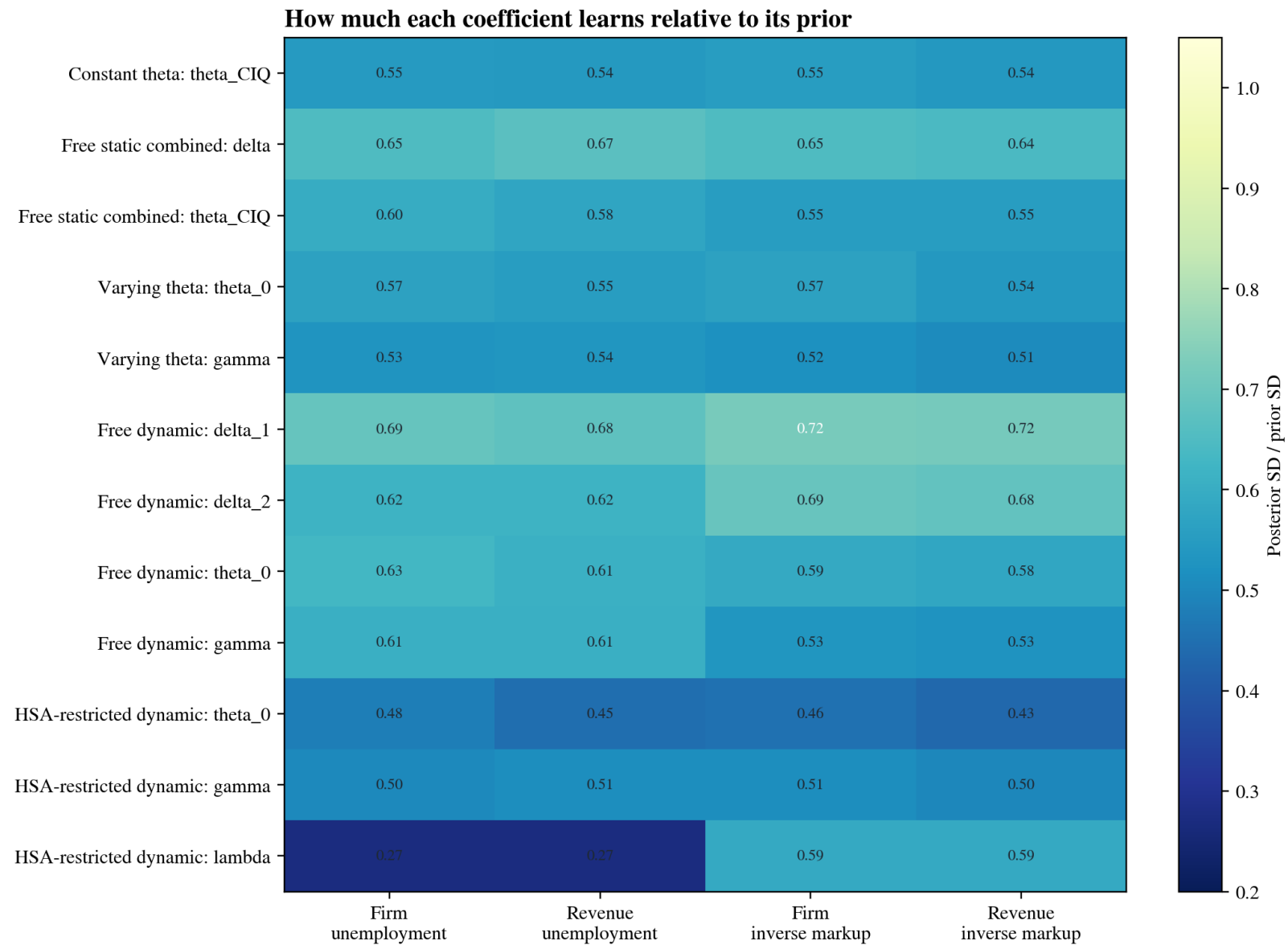


Figure 5: Posterior-to-prior SD ratios for target competition coefficients. Lower values mean more learning, not necessarily correct-sign identification.

12 Simulation Recovery Before Imposing HSA

Static recovery simulates the actual sample, activity series, inflation persistence, and paired competition states, injecting standardized effects

$$s_\theta = \theta SD(\hat{n})/SD(\pi), \quad s_\delta = \delta SD(\bar{n}^c x)/SD(\pi). \quad (22)$$

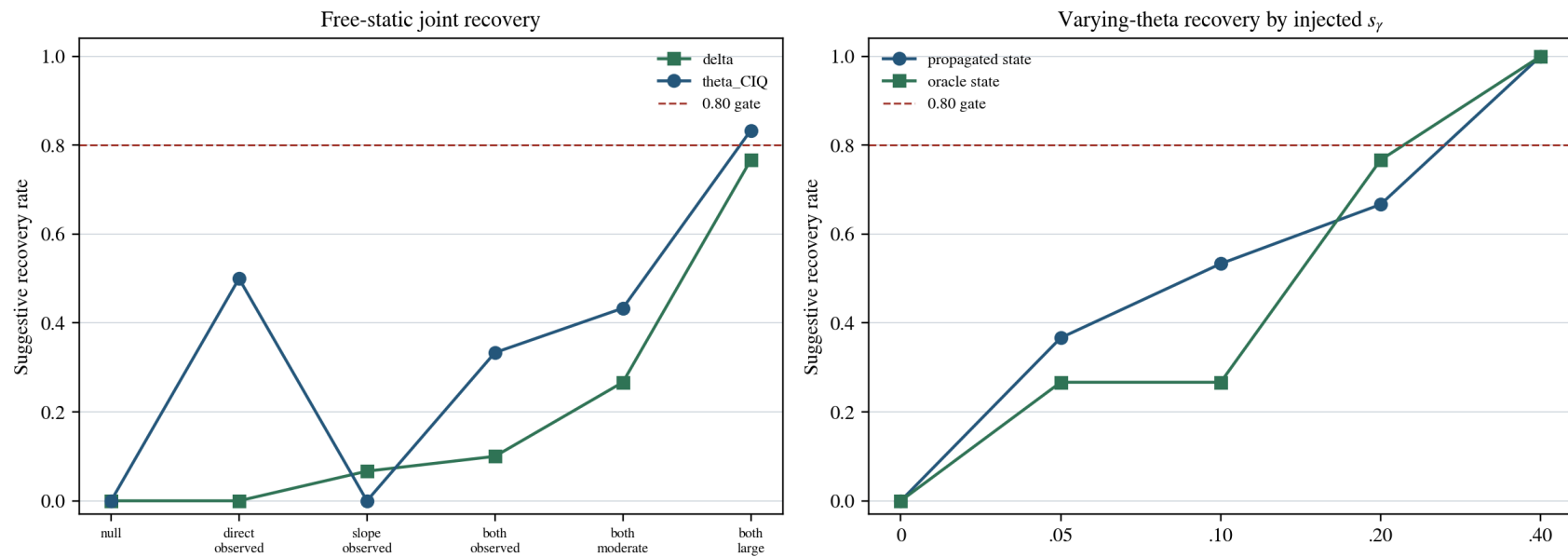
The observed unemployment-gap magnitudes are approximately $(s_\delta, s_\theta) = (0.06, 0.11)$.

Table 5: Primary propagated-state free-static recovery

Scenario	Parameter	True effect	Suggestive	Strong	Coverage
both_observed	delta	0.06	0.100	0.033	1.000
both_observed	theta_CIQ	0.11	0.333	0.067	1.000
both_moderate	delta	0.20	0.267	0.000	0.933
both_moderate	theta_CIQ	0.20	0.433	0.033	0.933
both_large	delta	0.40	0.767	0.033	0.800
both_large	theta_CIQ	0.40	0.833	0.233	0.700

At observed magnitudes, suggestive recovery is 0.10 for δ and 0.333 for theta, against the predeclared 0.80 promotion threshold. Oracle-state recovery is only 0.133 and 0.333. State measurement uncertainty is therefore not the main bottleneck; the aggregate design and effect sizes are.

The recovery computation itself passes: maximum R-hat is 1.0395, minimum bulk ESS is 100.6, coverage is 1.00 at the observed scenario, and null false-positive rates are zero. The gate fails because detection is weak, not because the simulation sampler failed.

Simulation recovery on the actual 83-quarter design**Figure 6:** Free-static and varying-theta recovery on the actual sample design.

13 M2 Results: Is Theta Time-Varying?

The isolated varying-theta equation is

$$\pi_t^q = a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q + \kappa_0 x_t - (\theta_0 + \gamma \bar{n}_t^c) \hat{n}_t + \varepsilon_t. \quad (23)$$

Table 6: Varying-theta model; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	-0.061 [-1.035, 0.920]	-0.119 [-1.112, 0.848]	-13.378 [-53.205, 26.472]	-14.644 [-55.328, 24.334]
theta_0	0.710 [-0.865, 2.262]	0.627 [-0.830, 2.077]	0.749 [-0.813, 2.283]	0.666 [-0.754, 2.099]
gamma	-0.165 [-1.030, 0.707]	-0.126 [-0.953, 0.701]	-0.215 [-1.083, 0.645]	-0.170 [-0.960, 0.610]

θ_0 remains directionally positive: $P(\theta_0 > 0)$ ranges from 0.804 to 0.830. Every interval still includes zero. More importantly, every γ posterior leans negative and includes zero; $P(\gamma > 0)$ ranges from 0.308 to 0.384. The data therefore do not reveal positive time variation in the direct channel.

The varying-theta recovery fixes $s_{\theta_0} = 0.11$ and injects $s_\gamma \in \{0, 0.05, 0.10, 0.20, 0.40\}$. Thirty propagated-state and thirty oracle-state replications are used at each value.

Table 7: Propagated-state recovery of γ

True s_γ	Suggestive	Strong	Coverage	Mean SD ratio
0.00	0.000	0.000	1.000	0.522
0.05	0.367	0.000	1.000	0.523
0.10	0.533	0.067	1.000	0.521
0.20	0.667	0.167	1.000	0.510
0.40	1.000	0.800	0.967	0.520

At $s_\gamma = 0.10$, suggestive recovery is 0.533 but strong recovery is only 0.067. Strong recovery reaches 0.80 only at $s_\gamma = 0.40$. Modest time variation is below the practical detection threshold of this aggregate sample.

14 M3 Results: Free Dynamic Channels

The unrestricted dynamic benchmark is

$$\theta_t = \theta_0 + \gamma \bar{n}_t^c, \quad (24)$$

$$\kappa_t = \kappa_0 + \delta_1 \bar{n}_t^c + \delta_2 q_t^{(2)}. \quad (25)$$

Table 8: Free-dynamic model; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.118 [-1.215, 1.451]	0.050 [-1.245, 1.309]	-16.294 [-64.777, 31.296]	-18.518 [-66.930, 31.734]
delta_1	0.102 [-0.615, 0.851]	0.115 [-0.595, 0.838]	-0.076 [-3.463, 3.262]	-0.130 [-3.383, 3.238]
delta_2	-0.177 [-1.156, 0.803]	-0.138 [-1.134, 0.830]	0.813 [-1.530, 3.137]	0.828 [-1.516, 3.191]
theta_0	0.894 [-0.845, 2.598]	0.750 [-0.857, 2.327]	0.613 [-0.993, 2.296]	0.518 [-0.988, 2.040]
gamma	-0.241 [-1.257, 0.746]	-0.171 [-1.104, 0.766]	-0.191 [-1.064, 0.693]	-0.157 [-0.957, 0.663]

All sixteen target intervals for $\delta_1, \delta_2, \theta_0, \gamma$ include zero. θ_0 keeps a positive direction, with $P(\theta_0 > 0)$ from 0.750 to 0.847, but γ remains negative-leaning. Neither the linear nor quadratic slope change is independently identified.

This model is essential even though it is weak. It shows what the data say before HSA links the channels. Any precision gain in the restricted model must be assessed against these wide free coefficients.

15 M4 Results: HSA-Restricted Dynamic Model

The fully restricted equation is

$$\pi_t^q = a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q \quad (26)$$

$$+ \left(\kappa_0 + \lambda \theta_0 \bar{n}_t^c + \frac{\lambda \gamma}{2} q_t^{(2)} \right) x_t \quad (27)$$

$$- (\theta_0 + \gamma \bar{n}_t^c) \hat{n}_t + \varepsilon_t. \quad (28)$$

Table 9: HSA-restricted dynamic model; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.106 [-1.247, 1.526]	0.051 [-1.250, 1.426]	-14.523 [-65.287, 33.928]	-15.291 [-66.327, 32.481]
theta_0	0.406 [-0.619, 2.035]	0.334 [-0.620, 1.837]	0.375 [-0.664, 1.832]	0.332 [-0.653, 1.678]
gamma	-0.090 [-0.949, 0.721]	-0.084 [-0.927, 0.719]	-0.175 [-0.996, 0.668]	-0.170 [-0.922, 0.639]
lambda	0.402 [-5.288, 5.952]	0.523 [-5.307, 6.244]	-0.411 [-13.298, 12.196]	-0.518 [-12.947, 12.155]
derived:delta_1	0.149 [-0.644, 1.033]	0.132 [-0.648, 1.013]	-0.028 [-3.901, 3.895]	0.022 [-3.796, 3.808]
derived:delta_2	-0.016 [-0.705, 0.615]	-0.010 [-0.691, 0.651]	0.427 [-1.181, 3.065]	0.396 [-1.204, 2.957]

Every λ interval spans zero and both signs:

$$\lambda_{\text{firm,unemp}} = 0.402[-5.288, 5.952], \quad \lambda_{\text{firm,markup}} = -0.411[-13.298, 12.196].$$

Every derived $\delta_1 = \lambda \theta_0$ and $\delta_2 = \lambda \gamma / 2$ interval also spans zero.

The narrower θ_0 posterior in the restricted parameterization is not independent direct-channel identification. The likelihood observes products such as $\lambda \theta_0$ and $\lambda \gamma$. When the unrestricted factors are near zero, many combinations of signs and magnitudes generate similar fitted values. This produces a weakly identified multiplicative geometry. The restriction shares information; it does not create new variation in the data.

16 Implied Time-Varying Coefficients and In-Sample Fit

Dynamic coefficients: primary firm-weighted unemployment-gap cell

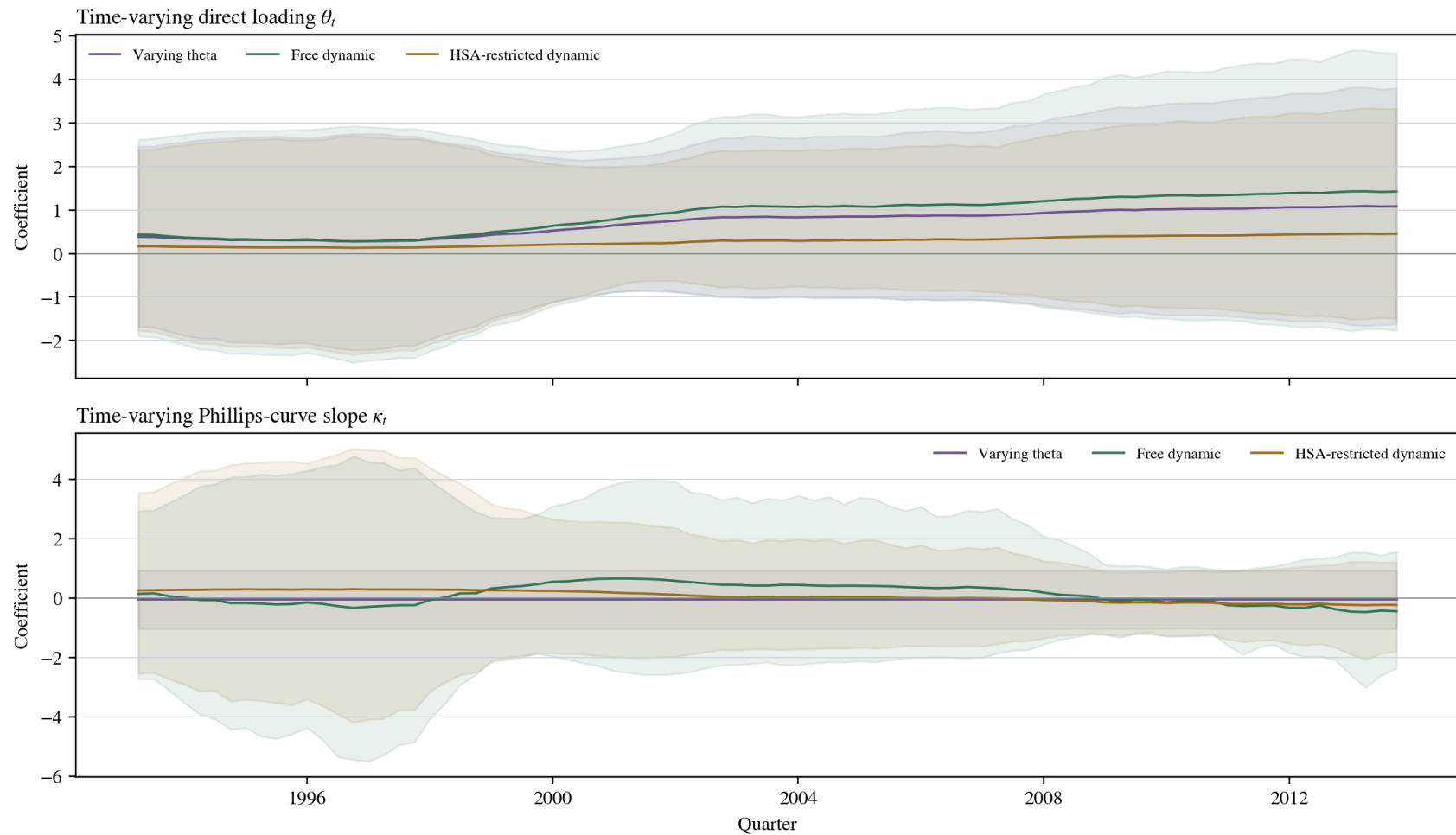


Figure 7: Posterior paths for the firm-weighted negative-unemployment-gap cell. Bands are pointwise 95% intervals.

The posterior medians move because they are functions of the estimated slow-state path. That visual movement must not be read as evidence that γ or the quadratic slope is identified. The bands remain wide and generally overlap zero. In particular, a smooth median path can be mechanically induced by multiplying an uncertain coefficient by a smooth $\bar{n}_t$.

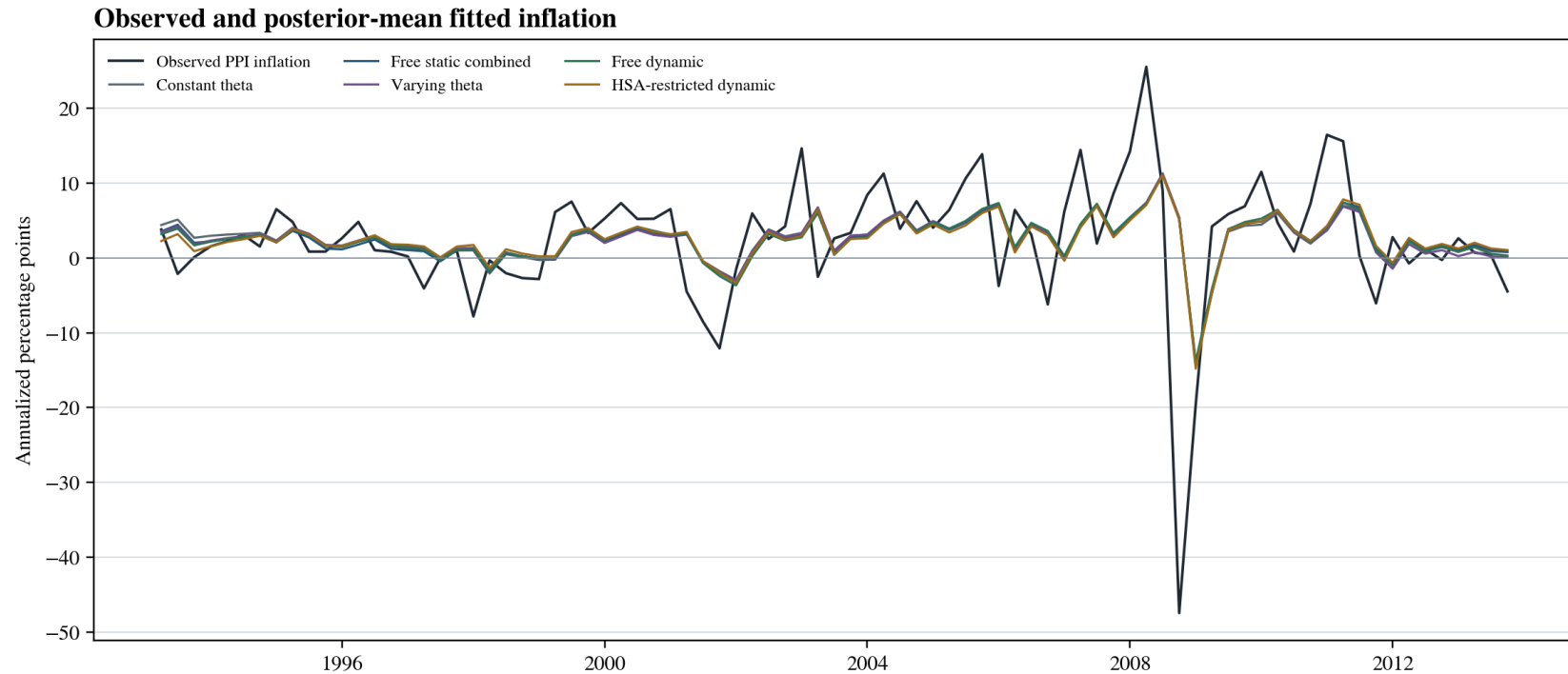


Figure 8: Observed PPI inflation and posterior-mean in-sample fitted values.

The fitted means are similar across specifications. Extra dynamic coefficients mostly redistribute a small fitted competition contribution rather than explaining the large QoQ inflation movements. This is consistent with the weak predictive gains reported next.

17 Model Comparison

The dynamic models are compared with the constant-theta baseline using full-sample PSIS-LOO, manually computed WAIC, and a fixed 2010Q1–2013Q4 holdout. Positive ELPD differences favor the dynamic model; negative RMSE differences favor it.

Table 10: Dynamic models relative to constant theta

Cycle	Activity	Model	Δ LOO	Δ WAIC	Δ holdout	Δ RMSE	max k
firm_weighted	inverse_markup	Varying theta	-0.088	-0.413	-0.828	-0.108	1.46
firm_weighted	inverse_markup	Free dynamic	-0.532	-0.531	-1.988	0.108	1.44
firm_weighted	inverse_markup	HSA-restricted dynamic	-0.944	-0.483	-1.145	-0.137	1.53
firm_weighted	negative_unemployment_gap	Varying theta	0.242	-0.522	-0.777	0.137	1.35
firm_weighted	negative_unemployment_gap	Free dynamic	-1.034	-1.752	-1.407	-0.111	1.42
firm_weighted	negative_unemployment_gap	HSA-restricted dynamic	-0.270	-1.213	-0.636	-0.274	1.58
revenue_weighted	inverse_markup	Varying theta	-0.031	-0.146	-1.341	-0.084	1.39
revenue_weighted	inverse_markup	Free dynamic	-0.135	-0.661	-2.444	0.082	1.43
revenue_weighted	inverse_markup	HSA-restricted dynamic	-0.925	-0.560	-1.603	-0.133	1.40
revenue_weighted	negative_unemployment_gap	Varying theta	-1.117	-0.736	-1.118	0.225	1.24
revenue_weighted	negative_unemployment_gap	Free dynamic	-2.195	-0.915	-1.850	0.012	1.44
revenue_weighted	negative_unemployment_gap	HSA-restricted dynamic	-1.167	-0.562	-0.774	-0.276	1.33

All twelve holdout ELPD differences are negative. Some dynamic models reduce holdout RMSE, but predictive density also evaluates uncertainty and tail behavior; those RMSE gains do not translate into better predictive distributions. LOO differences are small and unstable. Every comparison has at least one Pareto $k > 0.7$, with maxima above one, so PSIS-LOO is descriptive only.

Dynamic model comparison

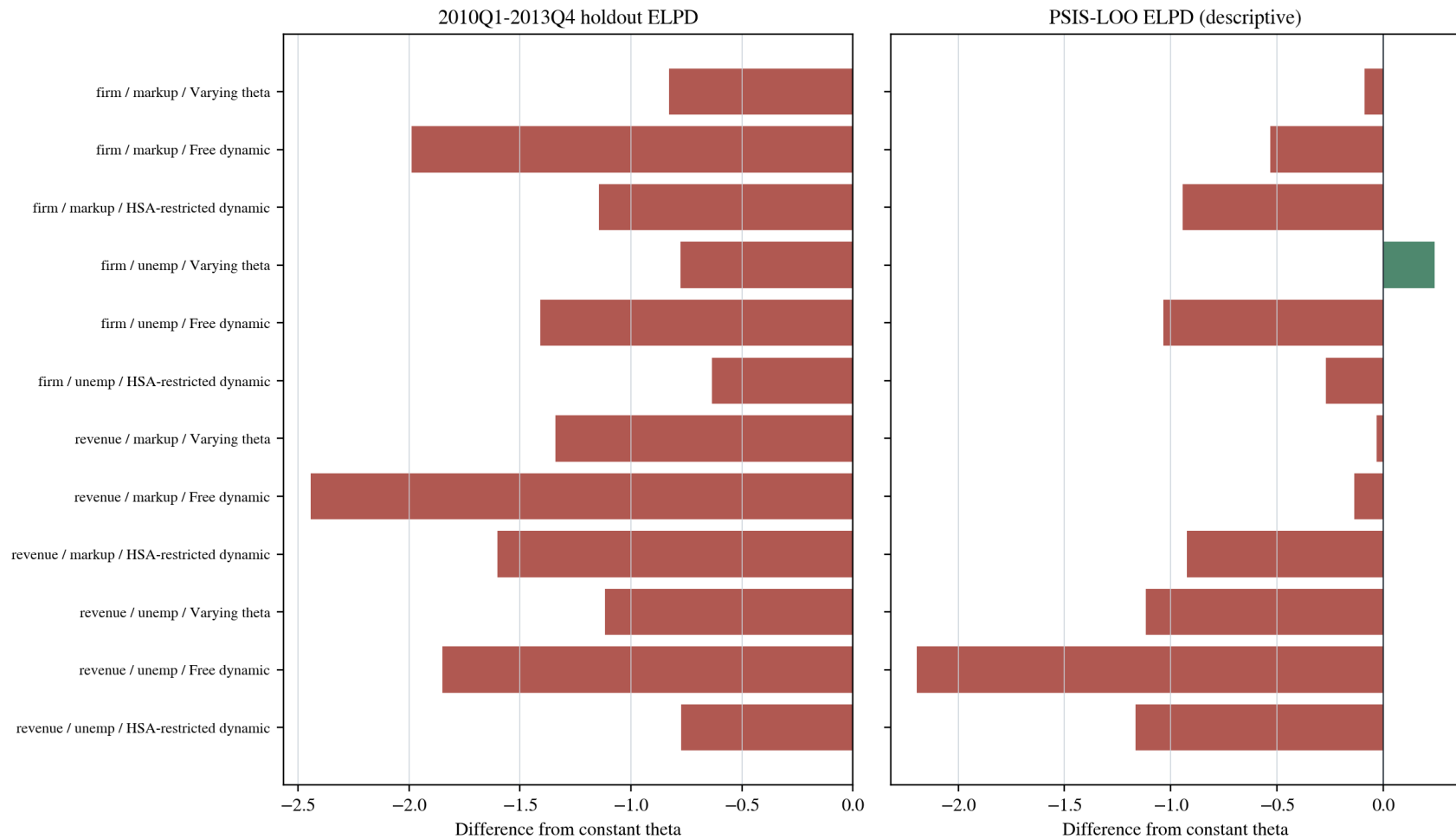


Figure 9: Predictive comparison relative to constant theta.

No formal marginal likelihood was computed for this post-gate dynamic diagnostic, and the current QoQ bundle did not re-estimate a separate CES model. Therefore this report does not make a dynamic-HSA-versus-CES Bayes-factor claim. Its valid nested comparison is free dynamic versus HSA restricted, with constant theta as the predictive baseline.

18 Identification Geometry

The relevant competition regressors are

$$-\hat{h}_t, \quad -\bar{n}_t^c \hat{h}_t, \quad \bar{n}_t^c x_t, \quad q_t^{(2)} x_t. \quad (29)$$

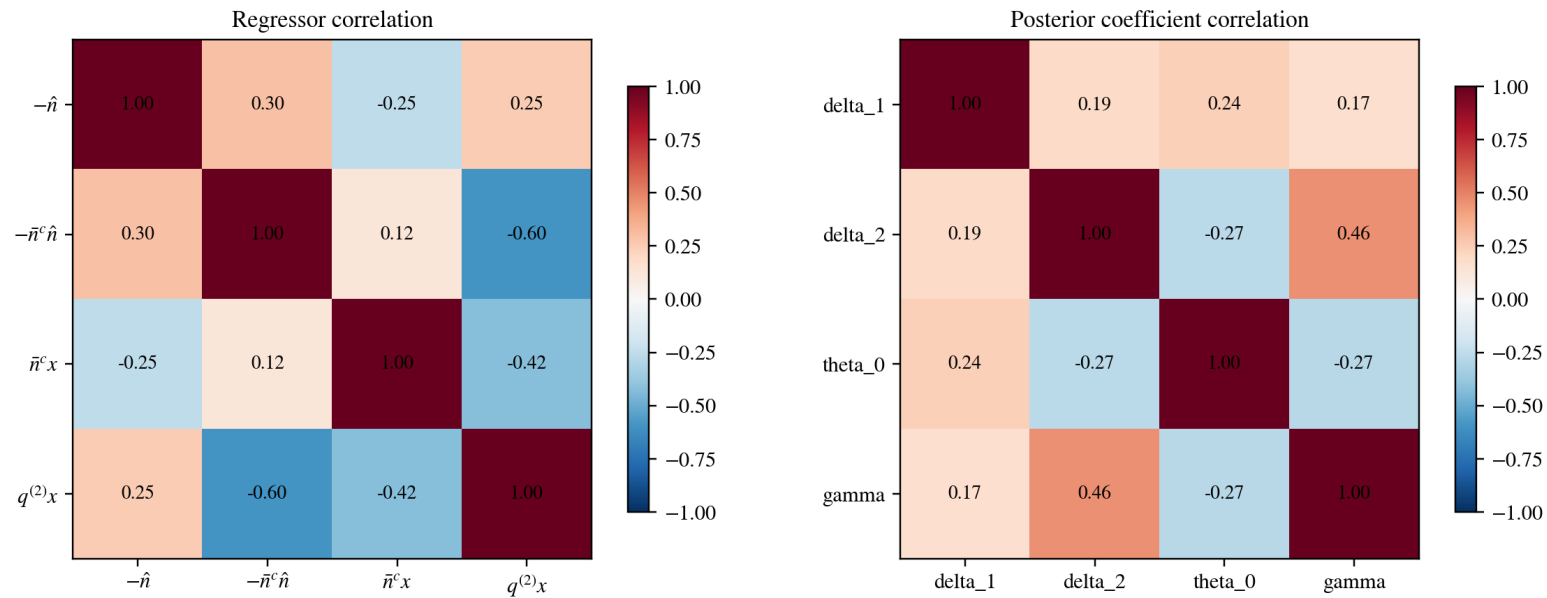
Their standardized condition numbers are not extreme: about 1.0–1.4 for varying theta and 2.3–3.0 for free dynamic. Maximum pairwise correlations are 0.42–0.68. Thus catastrophic linear collinearity is not the main explanation for failure.

Table 11: Regressor geometry using posterior-median competition paths

Cell	Corr(θ_0, γ regressors)	Corr(δ_1, δ_2 regressors)	Max regressor corr.	Cond. varying	Cond. free dyn.
Firm / unemployment	0.296	-0.423	0.601	1.36	3.03
Revenue / unemployment	-0.009	-0.393	0.676	1.01	2.96
Firm / inverse markup	0.296	0.388	0.416	1.36	2.34
Revenue / inverse markup	-0.009	0.402	0.497	1.01	2.36

Table 12: Posterior dependence among free-dynamic target coefficients

Cell	Max target posterior correlation
Firm / unemployment	0.456
Revenue / unemployment	0.481
Firm / inverse markup	0.375
Revenue / inverse markup	0.391

Identification geometry: firm-weighted unemployment-gap cell**Figure 10:** Regressor and posterior correlations in the primary firm-weighted unemployment-gap cell.

Posterior target correlations are moderate rather than singular. The problem is primarily low signal relative to quarterly inflation volatility, limited T , and the multiplicative HSA factorization—not a single near-collinear pair that could be removed to solve the model.

19 Persistent-Error Robustness and MCMC Convergence

Table 13: IID versus persistent AR(1), firm-weighted unemployment-gap cell

Model	Parameter	IID	Persistent AR(1)
Varying theta	theta_0	0.710 [-0.865, 2.262]	0.715 [-1.076, 2.543]
Varying theta	gamma	-0.165 [-1.030, 0.707]	-0.158 [-1.161, 0.847]
Free dynamic	delta_1	0.102 [-0.615, 0.851]	0.069 [-0.697, 0.832]
Free dynamic	delta_2	-0.177 [-1.156, 0.803]	-0.217 [-1.311, 0.866]
Free dynamic	theta_0	0.894 [-0.845, 2.598]	0.860 [-1.106, 2.782]
Free dynamic	gamma	-0.241 [-1.257, 0.746]	-0.231 [-1.338, 0.866]
HSA-restricted dynamic	theta_0	0.406 [-0.619, 2.035]	0.351 [-0.796, 2.089]
HSA-restricted dynamic	gamma	-0.090 [-0.949, 0.721]	-0.096 [-1.113, 0.847]
HSA-restricted dynamic	lambda	0.402 [-5.288, 5.952]	0.401 [-5.975, 6.292]

Allowing persistent inflation errors widens intervals modestly but does not change their sign conclusions. Free-dynamic γ is $-0.231[-1.338, 0.866]$ and HSA λ is $0.401[-5.975, 6.292]$ under AR(1).

Table 14: Observed IID fit convergence

Model	Cell	Max R-hat	Min bulk ESS
Constant theta	Firm / unemployment	1.0019	2466.9
Constant theta	Revenue / unemployment	1.0008	2390.1
Constant theta	Firm / inverse markup	1.0008	2711.9
Constant theta	Revenue / inverse markup	1.0019	2679.6
Free static combined	Firm / unemployment	1.0011	2514.6
Free static combined	Revenue / unemployment	1.0014	2725.2
Free static combined	Firm / inverse markup	1.0013	2592.0
Free static combined	Revenue / inverse markup	1.0024	2654.5
Varying theta	Firm / unemployment	1.0008	6488.6
Varying theta	Revenue / unemployment	1.0008	6832.8
Varying theta	Firm / inverse markup	1.0003	6357.6
Varying theta	Revenue / inverse markup	1.0007	6586.1
Free dynamic	Firm / unemployment	1.0003	6528.4
Free dynamic	Revenue / unemployment	1.0006	6615.1
Free dynamic	Firm / inverse markup	1.0004	6521.4
Free dynamic	Revenue / inverse markup	1.0005	6723.2
HSA-restricted dynamic	Firm / unemployment	1.0011	4569.7
HSA-restricted dynamic	Revenue / unemployment	1.0005	4729.5
HSA-restricted dynamic	Firm / inverse markup	1.0007	4273.2
HSA-restricted dynamic	Revenue / inverse markup	1.0006	4310.0

Every observed fit passes comfortably. Across the dynamic fits, maximum R-hat is 1.0011 and minimum bulk ESS is 4,074.5. Recovery maximum R-hat is 1.0129 and minimum bulk ESS is 498.5. The substantive failure is therefore not an MCMC convergence failure.

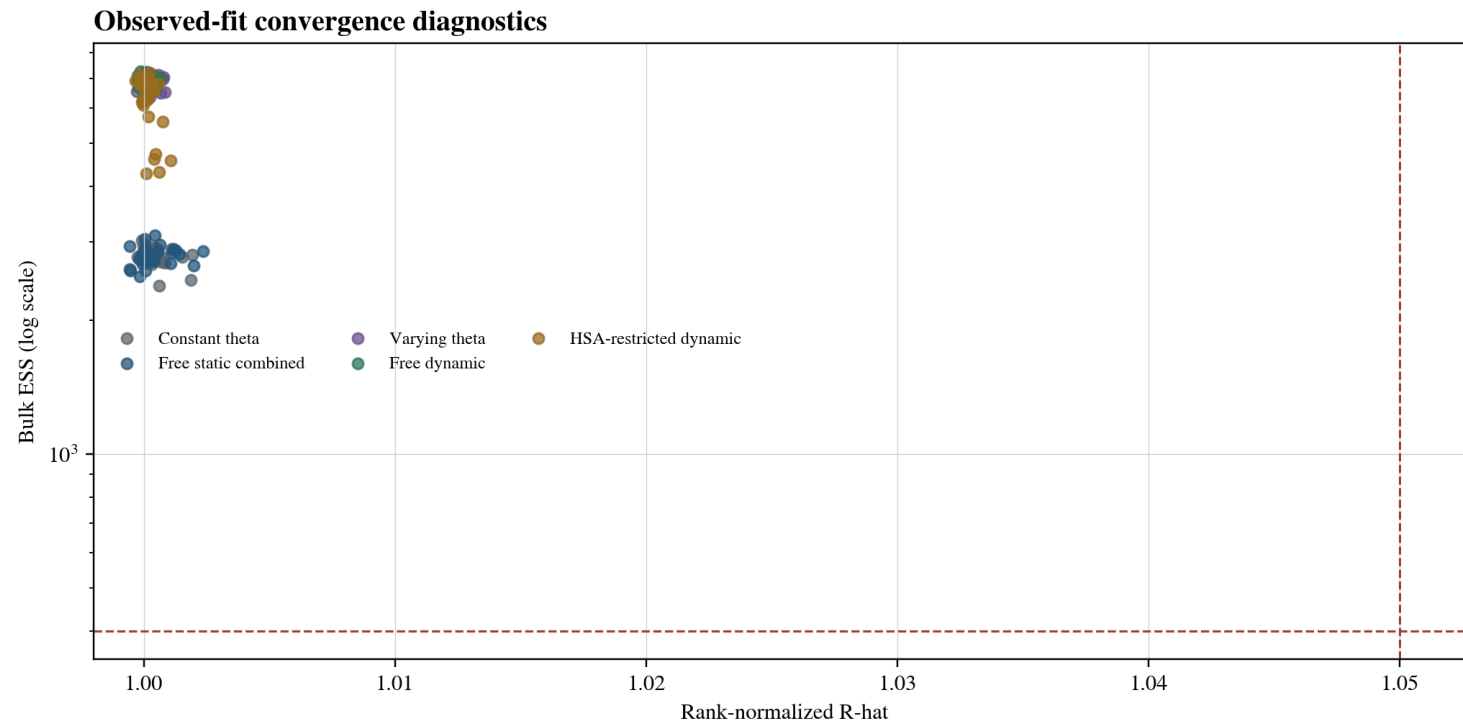


Figure 11: Parameter-level convergence diagnostics. Dashed lines are the dynamic-fit gates.

20 Core CPI Extension: Matched Data and Design

Core CPI is estimated as a separate outcome, not pooled with PPI. The competition-state draws, 1993Q2–2013Q4 sample, activity variables, priors, M0–M4 ladder, IID primary error, AR(1) robustness, and 2010Q1 holdout are held fixed. The dependent variable is

$$\pi_t^{q,core} = 400 (\log P_t^{core} - \log P_{t-1}^{core}), \quad (30)$$

where monthly FRED CPILFESL is averaged within quarter before differencing.

The expectation is the Philadelphia Fed SPF mean headline-CPI forecast CPI3. At survey quarter t , CPI3 is the genuine forecast of annualized headline-CPI inflation in $t + 1$, converted to $100 \log(1 + r_t/100)$. It is horizon-matched and closer to Core CPI than the GDP-price forecast used for PPI, but it remains a *headline-CPI* proxy because an independent historical Core-CPI expectation is unavailable.

Table 15: Core-CPI QoQ data summary

Series	Source	Transformation	Start	End	T	Mean	SD
Core CPI inflation	FRED CPILFESL	$400\Delta \log P_t$	1993Q2	2013Q4	83	2.155	0.644
SPF headline-CPI expectation	Philadelphia Fed CPI3	$100 \log(1 + r_t/100)$	1993Q2	2013Q4	83	2.304	0.571
Negative unemployment gap	FRED NROU and UNRATE	$u_t^* - u_t$	1993Q2	2013Q4	83	-0.940	1.738
Inverse-markup proxy	Nekarda-Ramey markup	$\log(1.2/\mu_t)$	1993Q2	2013Q4	83	-0.333	0.038

PPI and Core CPI observables on the matched 1993Q2-2013Q4 sample

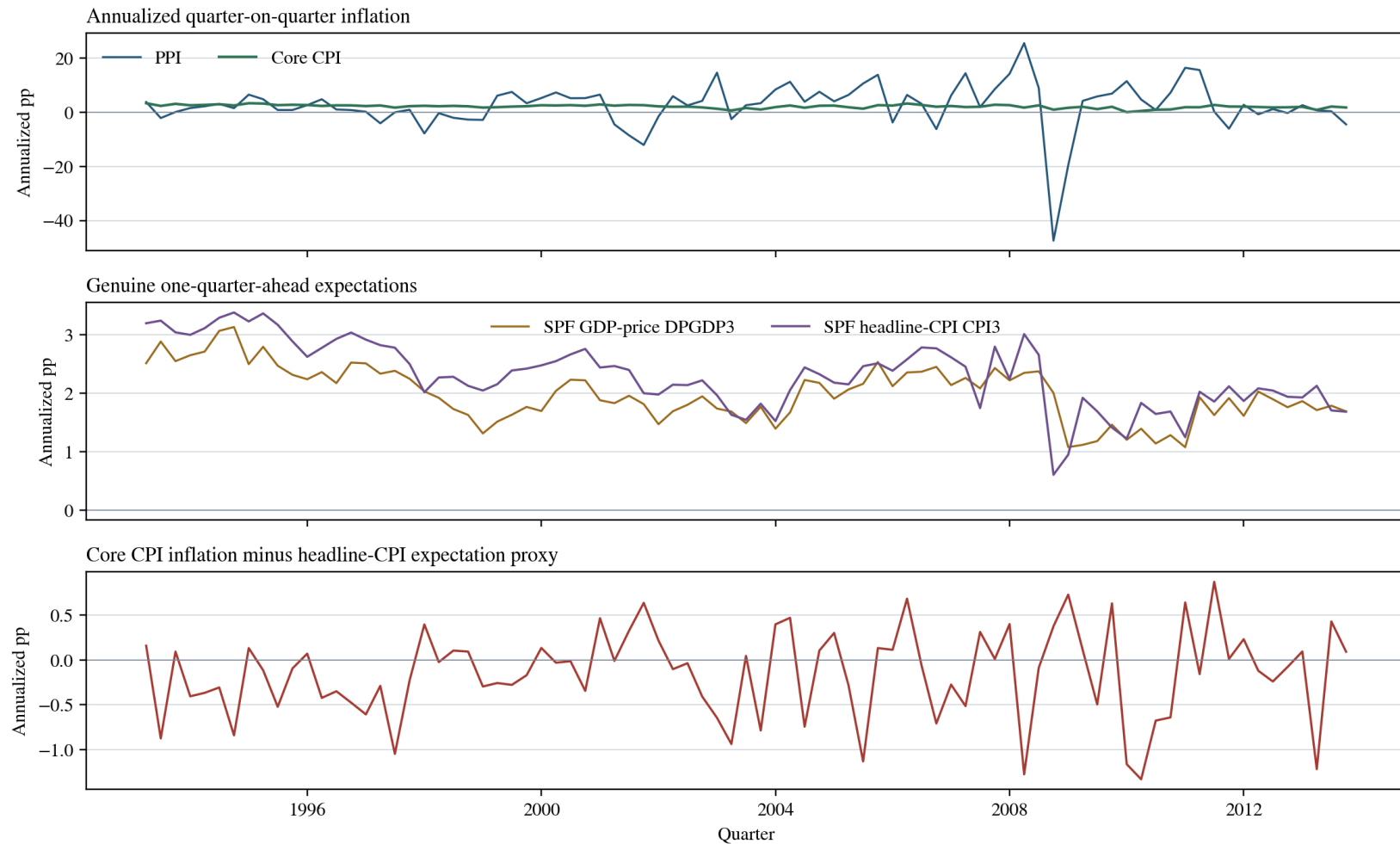


Figure 12: Matched PPI and Core-CPI outcomes and their price-appropriate SPF proxies.

Core CPI is much smoother than PPI. This changes the signal-to-noise environment and prior scales, but it does not add observations: both equations still contain only 83 quarters. The exercise therefore asks whether the PPI direction transports to a less volatile consumer price outcome, not whether the two inflation series share coefficients.

21 Core CPI M0 and M1: Average Direct and Free Static Channels

For M0 the Core-CPI equation is

$$\pi_t^{q,core} = a + \alpha_b \pi_{t-1}^{q,core} + \alpha_f E_t \pi_{t+1}^{q,CPI} + \kappa_0 x_t - \theta_{CIQ} \hat{n}_t + \varepsilon_t. \quad (31)$$

Table 16: Core CPI M0 constant direct loading; posterior mean and 95% interval

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
alpha_b	0.201 [-0.003, 0.406]	0.200 [-0.012, 0.398]	0.209 [-0.006, 0.412]	0.204 [0.002, 0.405]
alpha_f	0.546 [0.314, 0.780]	0.546 [0.300, 0.789]	0.570 [0.335, 0.804]	0.576 [0.338, 0.815]
kappa_0	0.042 [-0.022, 0.106]	0.043 [-0.022, 0.108]	1.372 [-1.225, 3.908]	1.415 [-1.275, 3.941]
theta_CIQ	-0.016 [-0.111, 0.078]	-0.016 [-0.101, 0.069]	-0.025 [-0.118, 0.067]	-0.024 [-0.112, 0.063]

Unlike PPI, every Core-CPI M0 posterior mean for θ_{CIQ} is negative. Across the four cycle/activity cells, $P(\theta_{CIQ} > 0)$ is only 0.294–0.378. The posterior SD is 43–46% of its prior SD, so the sign reversal is not caused by a failure to update. Yet all four intervals include zero; the result is directional evidence against transportability, not a sign-identified negative structural effect.

M1 adds the slow-state slope without linking it to the direct loading:

$$\pi_t^{q,core} = \dots + (\kappa_0 + \delta \bar{n}_t^c) x_t - \theta_{CIQ} \hat{n}_t + \varepsilon_t. \quad (32)$$

Table 17: Core CPI M1 free-static combined channels

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.024 [-0.060, 0.113]	0.024 [-0.058, 0.105]	1.415 [-1.576, 4.273]	1.438 [-1.455, 4.404]
delta	-0.016 [-0.067, 0.037]	-0.019 [-0.071, 0.033]	0.009 [-0.227, 0.241]	0.007 [-0.218, 0.237]
theta_CIQ	-0.029 [-0.133, 0.072]	-0.032 [-0.129, 0.069]	-0.026 [-0.124, 0.077]	-0.023 [-0.114, 0.069]

Adding δ makes the direct mean slightly more negative. In unemployment-gap cells, δ also leans negative; in inverse-markup cells it is essentially centered on zero. Every interval includes zero. Thus M1 does not reveal a hidden positive Core-CPI direct channel, nor does it identify the standalone slow-slope channel.

22 Core CPI M2: Does the Direct Loading Vary?

The isolated time-variation diagnostic is

$$\pi_t^{q,core} = \dots + \kappa_0 x_t - (\theta_0 + \gamma \bar{n}_t^c) \hat{n}_t + \varepsilon_t. \quad (33)$$

Table 18: Core CPI M2 varying-theta model

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.056 [-0.013, 0.125]	0.057 [-0.015, 0.129]	1.647 [-1.120, 4.385]	1.725 [-0.950, 4.535]
theta_0	-0.026 [-0.121, 0.069]	-0.016 [-0.104, 0.073]	-0.035 [-0.134, 0.060]	-0.023 [-0.113, 0.066]
gamma	0.030 [-0.025, 0.087]	0.025 [-0.027, 0.078]	0.022 [-0.032, 0.077]	0.018 [-0.032, 0.067]

θ_0 retains the negative M0 direction. By contrast, every γ mean is positive. The strongest directional cell is firm-weighted Core CPI $\times$ unemployment gap, $\gamma = 0.030[-0.025, 0.087]$ with $P(\gamma > 0) = 0.860$. The revenue-weighted version gives $0.025[-0.027, 0.078]$ with probability 0.829. This pattern is worth reporting, but it does not meet interval or recovery criteria and therefore cannot be described as identified time variation.

23 Core CPI M3 and M4: Free Dynamic versus HSA Dynamic

The unrestricted M3 benchmark estimates

$$\theta_t = \theta_0 + \gamma \bar{n}_t^c, \quad \kappa_t = \kappa_0 + \delta_1 \bar{n}_t^c + \delta_2 q_t^{(2)}. \quad (34)$$

Table 19: Core CPI M3 free-dynamic model

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.039 [-0.053, 0.128]	0.037 [-0.054, 0.129]	1.233 [-2.147, 4.604]	1.234 [-2.265, 4.732]
delta_1	-0.023 [-0.081, 0.034]	-0.024 [-0.081, 0.032]	-0.038 [-0.306, 0.223]	-0.044 [-0.310, 0.217]
delta_2	-0.024 [-0.097, 0.048]	-0.020 [-0.096, 0.056]	0.040 [-0.117, 0.201]	0.043 [-0.124, 0.209]
theta_0	-0.028 [-0.138, 0.088]	-0.025 [-0.128, 0.079]	-0.042 [-0.143, 0.060]	-0.031 [-0.126, 0.069]
gamma	0.018 [-0.048, 0.083]	0.015 [-0.047, 0.078]	0.026 [-0.030, 0.082]	0.020 [-0.033, 0.072]

All M3 intervals for δ_1 , δ_2 , θ_0 , and γ include zero. The markup cells lean toward $\delta_2 > 0$ and $\gamma > 0$, while the unemployment-gap cells lean toward negative slope changes. These differences show why a restriction should not be credited with identification before the free coefficients are recovered.

M4 imposes the HSA cross-equation restriction:

$$\pi_t^{q,core} = a + \alpha_b \pi_{t-1}^{q,core} + \alpha_f E_t \pi_{t+1}^{q,CPI} \quad (35)$$

$$+ \left(\kappa_0 + \lambda \theta_0 \bar{n}_t^c + \frac{\lambda \gamma}{2} q_t^{(2)} \right) x_t \quad (36)$$

$$- (\theta_0 + \gamma \bar{n}_t^c) \hat{n}_t + \varepsilon_t. \quad (37)$$

Table 20: Core CPI M4 HSA-restricted dynamic model

Parameter	Firm / unemployment	Revenue / unemployment	Firm / inverse markup	Revenue / inverse markup
kappa_0	0.030 [-0.074, 0.125]	0.029 [-0.074, 0.120]	1.174 [-2.651, 4.617]	1.265 [-2.422, 4.602]
theta_0	-0.016 [-0.111, 0.044]	-0.015 [-0.105, 0.046]	-0.028 [-0.112, 0.047]	-0.019 [-0.093, 0.051]
gamma	0.009 [-0.057, 0.071]	0.008 [-0.055, 0.067]	0.023 [-0.029, 0.080]	0.019 [-0.029, 0.070]
lambda	0.564 [-8.287, 9.120]	0.775 [-7.771, 9.291]	1.056 [-10.042, 11.710]	1.185 [-10.358, 12.916]
derived:delta_1	-0.027 [-0.114, 0.031]	-0.027 [-0.109, 0.033]	-0.054 [-0.429, 0.243]	-0.049 [-0.422, 0.225]
derived:delta_2	-0.019 [-0.112, 0.030]	-0.015 [-0.108, 0.036]	0.028 [-0.096, 0.214]	0.028 [-0.091, 0.228]

Every λ interval spans both signs. In the firm-weighted unemployment-gap cell, $\lambda = 0.564[-8.287, 9.120]$; in the corresponding inverse-markup cell it is $1.056[-10.042, 11.710]$. All derived $\delta_1 = \lambda\theta_0$ and $\delta_2 = \lambda\gamma/2$ intervals include zero. As in PPI, posterior narrowing inside M4 reflects shared multiplicative restrictions and is not independent evidence for the direct channel.

24 PPI versus Core CPI: Coefficients on a Common Layout

The following table holds the Capital IQ aggregation fixed at firm weighting and places the two inflation outcomes and two activity measures in columns. The equations are exactly M0–M4 above; only π_t , its lag, and its expectation proxy change across price indices.

Table 21: Firm-weighted target coefficients across PPI and Core CPI

Target	PPI / unemployment	PPI / inverse markup	Core CPI / unemployment	Core CPI / inverse markup
M0 θ	0.674 [-0.845, 2.200]	0.656 [-0.843, 2.176]	-0.016 [-0.111, 0.078]	-0.025 [-0.118, 0.067]
M1 δ	0.173 [-0.517, 0.861]	0.221 [-2.779, 3.255]	-0.016 [-0.067, 0.037]	0.009 [-0.227, 0.241]
M1 θ	0.784 [-0.832, 2.388]	0.672 [-0.874, 2.189]	-0.029 [-0.133, 0.072]	-0.026 [-0.124, 0.077]
M2 θ_0	0.710 [-0.865, 2.262]	0.749 [-0.813, 2.283]	-0.026 [-0.121, 0.069]	-0.035 [-0.134, 0.060]
M2 γ	-0.165 [-1.030, 0.707]	-0.215 [-1.083, 0.645]	0.030 [-0.025, 0.087]	0.022 [-0.032, 0.077]
M3 δ_1	0.102 [-0.615, 0.851]	-0.076 [-3.463, 3.262]	-0.023 [-0.081, 0.034]	-0.038 [-0.306, 0.223]
M3 δ_2	-0.177 [-1.156, 0.803]	0.813 [-1.530, 3.137]	-0.024 [-0.097, 0.048]	0.040 [-0.117, 0.201]
M3 θ_0	0.894 [-0.845, 2.598]	0.613 [-0.993, 2.296]	-0.028 [-0.138, 0.088]	-0.042 [-0.143, 0.060]
M3 γ	-0.241 [-1.257, 0.746]	-0.191 [-1.064, 0.693]	0.018 [-0.048, 0.083]	0.026 [-0.030, 0.082]
M4 θ_0	0.406 [-0.619, 2.035]	0.375 [-0.664, 1.832]	-0.016 [-0.111, 0.044]	-0.028 [-0.112, 0.047]
M4 γ	-0.090 [-0.949, 0.721]	-0.175 [-0.996, 0.668]	0.009 [-0.057, 0.071]	0.023 [-0.029, 0.080]
M4 λ	0.402 [-5.288, 5.952]	-0.411 [-13.298, 12.196]	0.564 [-8.287, 9.120]	1.056 [-10.042, 11.710]
M4 δ_1	0.149 [-0.644, 1.033]	-0.028 [-3.901, 3.895]	-0.027 [-0.114, 0.031]	-0.054 [-0.429, 0.243]
M4 δ_2	-0.016 [-0.705, 0.615]	0.427 [-1.181, 3.065]	-0.019 [-0.112, 0.030]	0.028 [-0.096, 0.214]

The comparison makes the central result visible. PPI M0/M1 θ means are positive, whereas Core-CPI means are negative. PPI γ leans negative, whereas Core-CPI γ leans positive. No corresponding 95% interval excludes zero. Consequently the cross-price exercise weakens, rather than strengthens, a claim of a common HSA coefficient. Pooling the two price systems would conceal this economically relevant heterogeneity.

25 Core CPI Prior–Posterior Learning and Recovery

Core CPI prior versus posterior: firm-weighted unemployment-gap cell

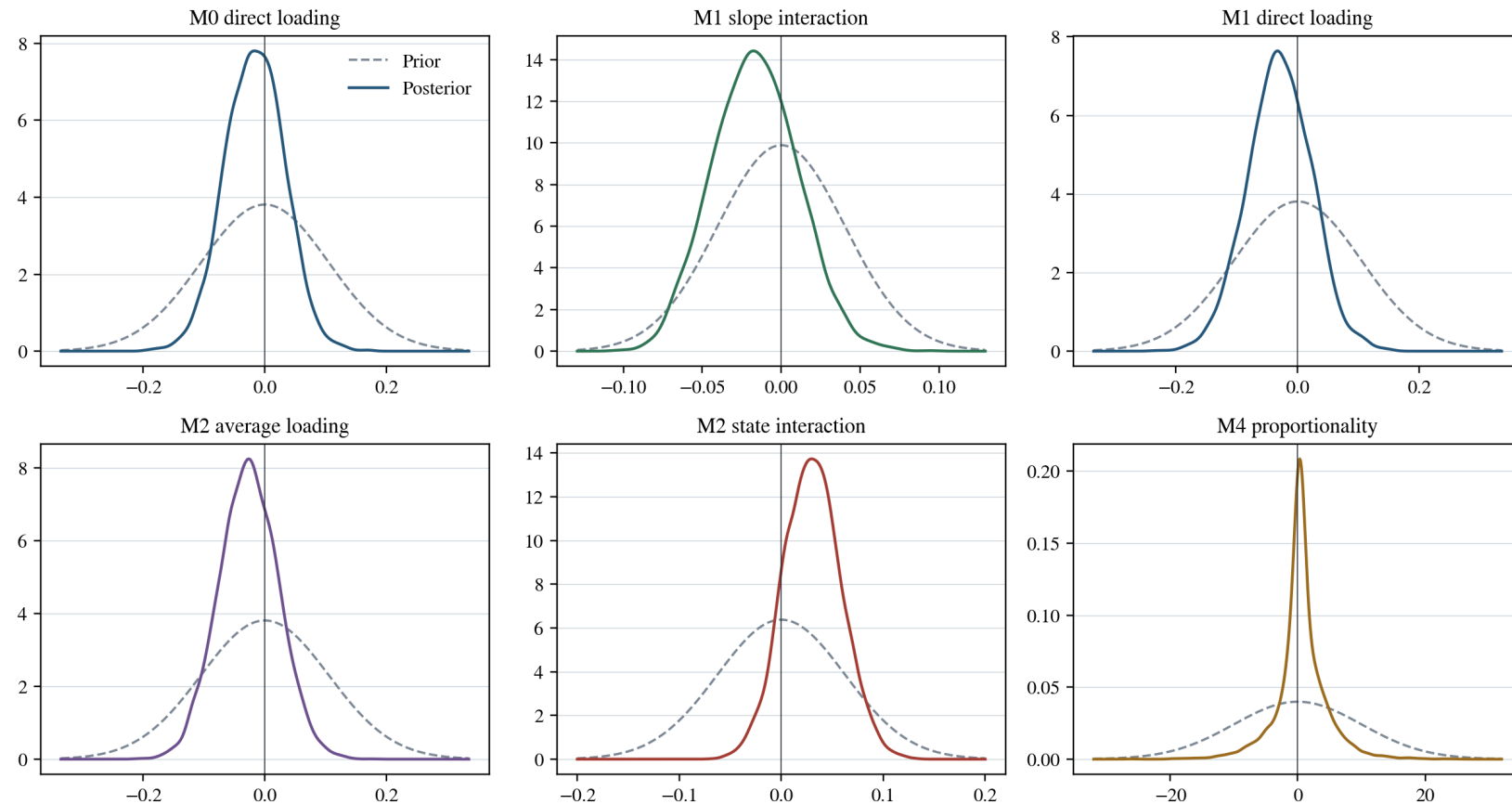


Figure 13: Core CPI selected prior-versus-posterior densities.

The posteriors narrow substantially relative to their standardized priors. This is genuine Bayesian learning about the empirical regressions. It is nevertheless distinct from learning the theoretically maintained sign: a narrow posterior can lie mostly on the opposite side of zero, and every reported interval can still span zero.

The Core-CPI recovery experiment repeats the PPI grids on the actual Core-CPI outcome, activity paths, persistence, and frozen state draws. It contains 480 static and 300 dynamic fits, with 30 propagated-state and 30 oracle-state replications per injected effect.

Table 22: Core CPI propagated-state free-static recovery

Scenario	Parameter	True effect	Suggestive	Strong	Coverage
both_observed	delta	0.06	0.000	0.000	1.000
both_observed	theta_CIQ	0.11	0.433	0.067	0.967
both_moderate	delta	0.20	0.200	0.000	0.867
both_moderate	theta_CIQ	0.20	0.767	0.033	0.933
both_large	delta	0.40	0.833	0.033	0.567
both_large	theta_CIQ	0.40	0.867	0.500	0.567

At $(s_\delta, s_\theta) = (0.06, 0.11)$, suggestive recovery is 0 for δ and 0.433 for θ . Oracle-state rates are 0.067 and 0.500. At $s_\gamma = 0.10$, propagated-state suggestive recovery is 0.367 and strong recovery is 0.067.

Table 23: Core CPI propagated-state recovery of γ

True s_γ	Suggestive	Strong	Coverage	Mean SD ratio
0.00	0.000	0.000	0.967	0.438
0.05	0.200	0.000	1.000	0.436
0.10	0.367	0.067	0.933	0.449
0.20	0.867	0.367	0.967	0.460
0.40	1.000	0.767	0.733	0.485

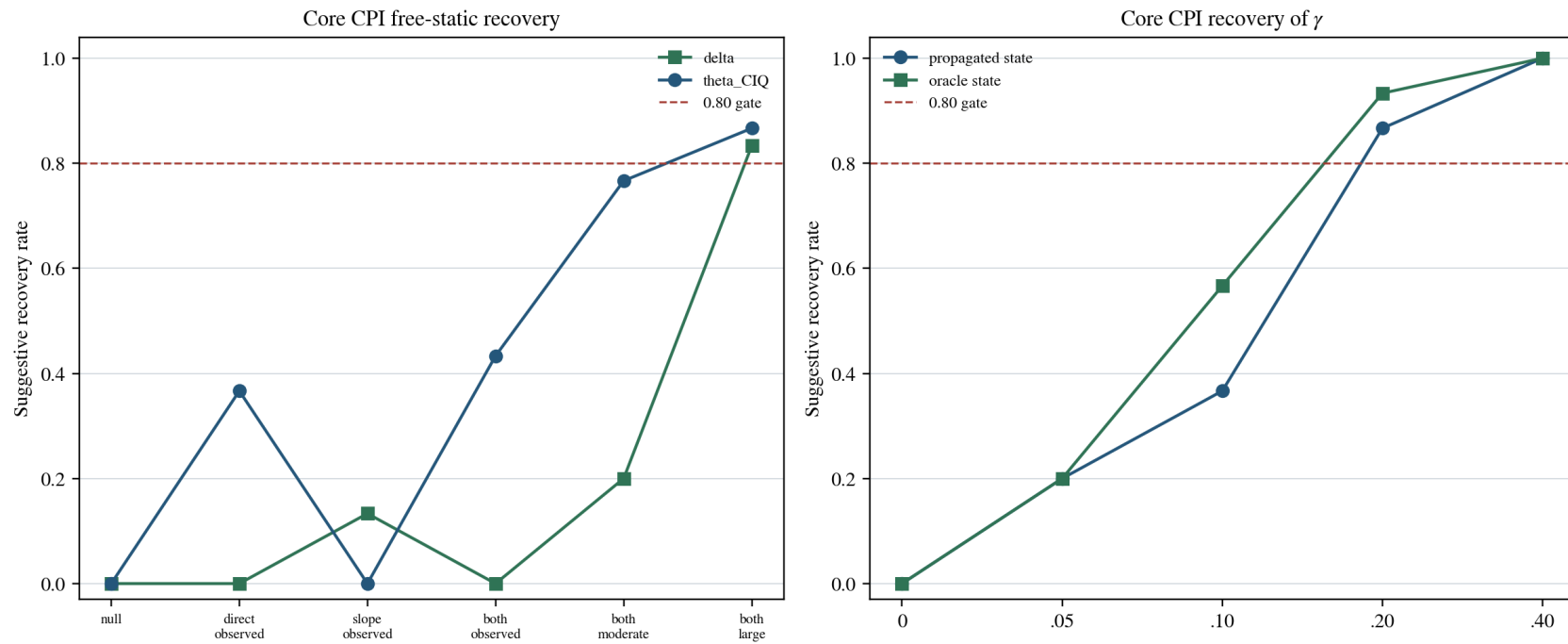


Figure 14: Core CPI free-static and varying-theta recovery.

The recovery sampler passes comfortably: maximum R-hat is 1.0207 and minimum bulk ESS is 331.9. The observed-scale effects fail because they are detected too infrequently, not because recovery chains fail to mix.

26 Core CPI Implied Paths and In-Sample Fit

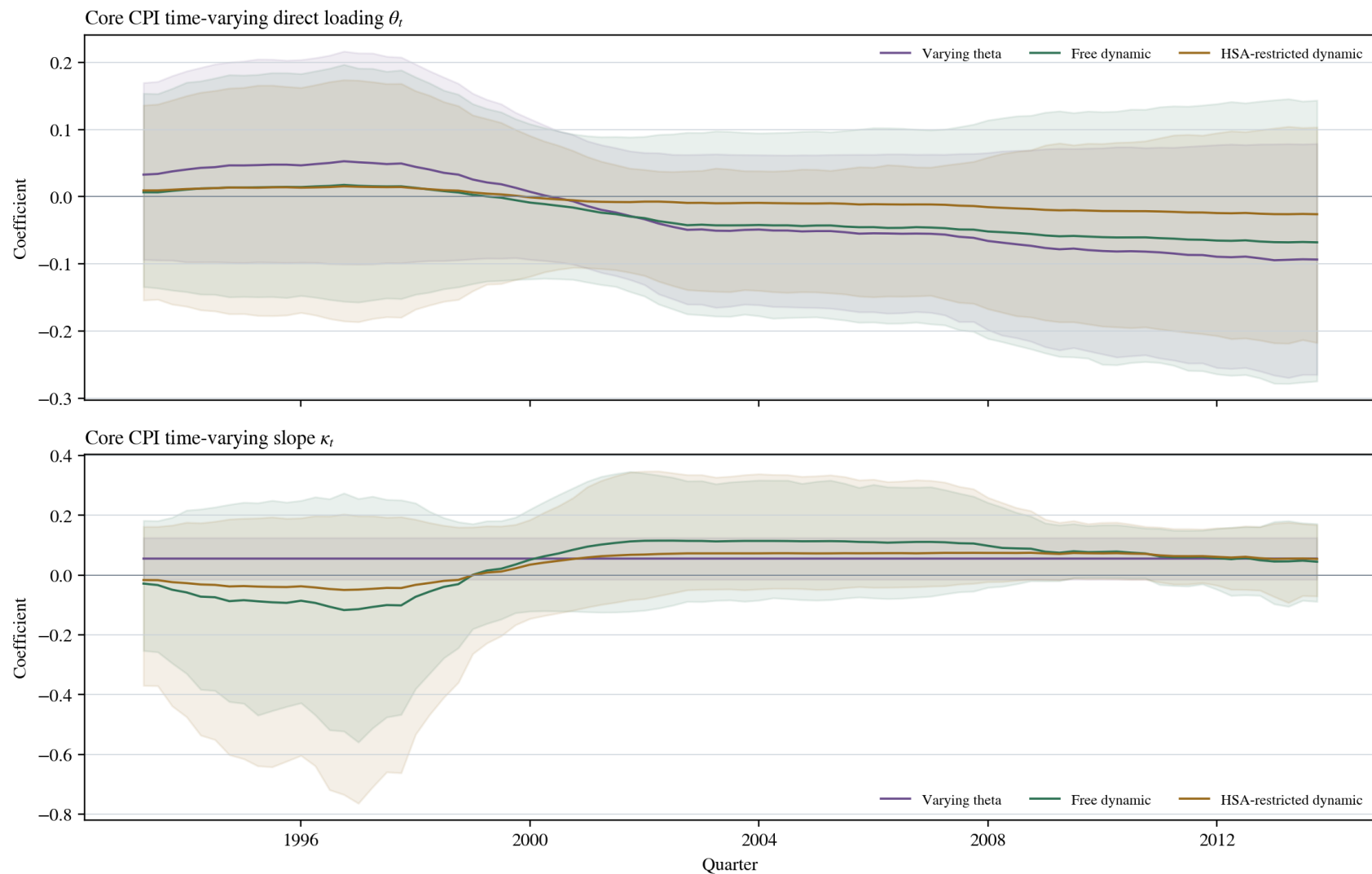


Figure 15: Core CPI implied θ_t and κ_t paths in the firm-weighted unemployment-gap cell.

The positive median drift in θ_t is generated by a positive but zero-crossing γ multiplied by the smooth slow state. The pointwise bands overlap zero over the sample. As for PPI, motion in the posterior median is not equivalent to identification of historical time variation.

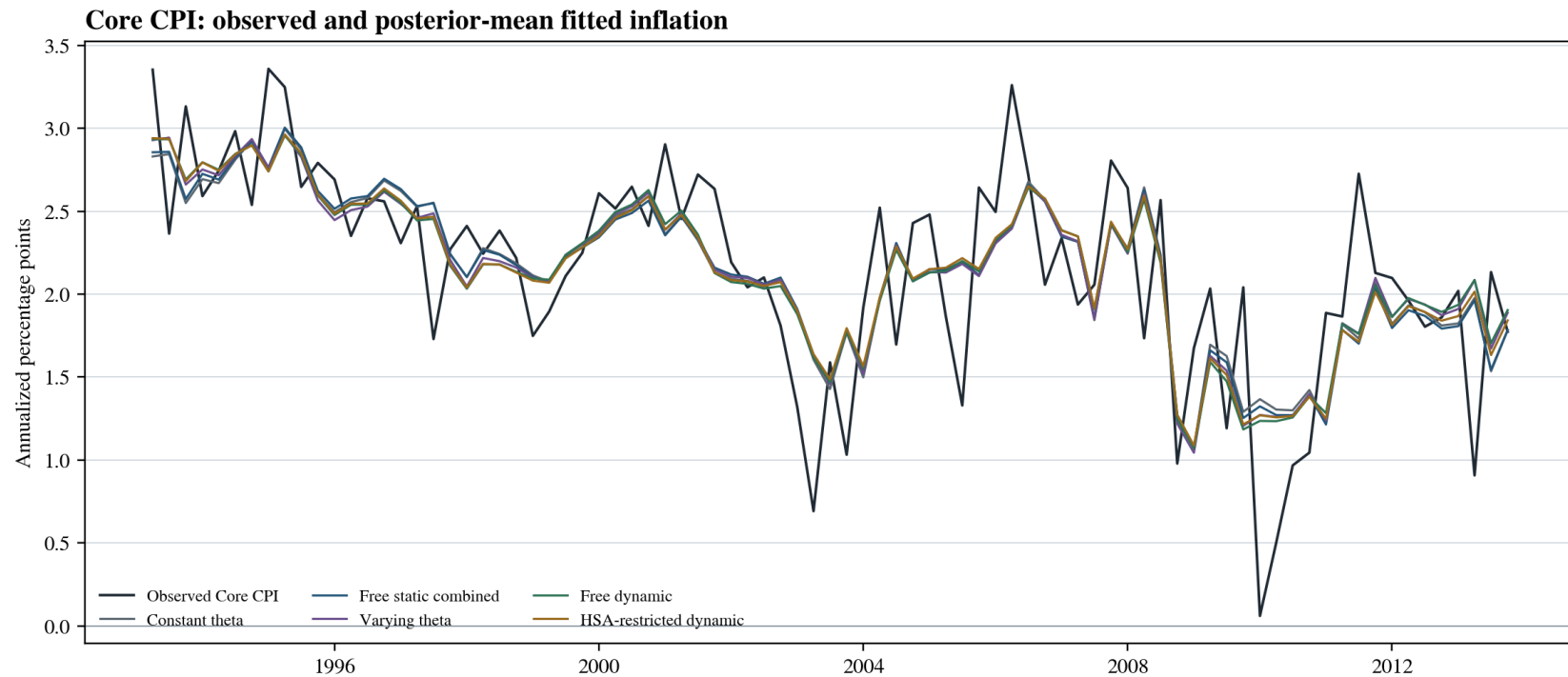


Figure 16: Observed Core CPI and posterior-mean fitted values.

The five fitted means are close. The extra competition interactions alter a small component of fitted Core CPI rather than accounting for a visibly distinct inflation movement.

27 Core CPI Prediction, AR(1) Robustness, and Convergence

Table 24: Core CPI M1–M4 relative to M0 constant theta

Cycle	Activity	Model	Δ LOO	Δ WAIC	Δ holdout	Δ RMSE	max k
firm_weighted	inverse_markup	Free static combined	-0.436	-0.459	0.124	-0.001	0.46
firm_weighted	inverse_markup	Varying theta	-0.240	-0.217	-0.443	0.017	0.51
firm_weighted	inverse_markup	Free dynamic	-1.046	-0.954	-0.092	0.009	0.55
firm_weighted	inverse_markup	HSA-restricted dynamic	-0.459	-0.415	0.036	0.006	0.47
firm_weighted	negative_unemployment_gap	Free static combined	-0.105	-0.133	0.321	-0.000	0.43
firm_weighted	negative_unemployment_gap	Varying theta	0.004	0.068	0.005	0.007	0.50
firm_weighted	negative_unemployment_gap	Free dynamic	-0.447	-0.315	-3.118	0.103	0.64
firm_weighted	negative_unemployment_gap	HSA-restricted dynamic	-0.236	-0.146	-2.348	0.079	0.44
revenue_weighted	inverse_markup	Free static combined	-0.268	-0.244	-0.037	0.002	0.46
revenue_weighted	inverse_markup	Varying theta	-0.563	-0.443	-0.769	0.024	0.55
revenue_weighted	inverse_markup	Free dynamic	-1.215	-1.095	-0.301	0.009	0.69
revenue_weighted	inverse_markup	HSA-restricted dynamic	-0.542	-0.480	-0.137	0.008	0.54
revenue_weighted	negative_unemployment_gap	Free static combined	-0.064	-0.060	0.309	-0.000	0.55
revenue_weighted	negative_unemployment_gap	Varying theta	-0.374	-0.299	-0.076	0.009	0.65
revenue_weighted	negative_unemployment_gap	Free dynamic	-0.740	-0.599	-3.326	0.110	0.56
revenue_weighted	negative_unemployment_gap	HSA-restricted dynamic	-0.433	-0.335	-2.211	0.078	0.57

Free static improves holdout ELPD slightly in three of four cells, by at most 0.321, while its LOO and WAIC differences are negative. The isolated M2 improvement is essentially zero in the firm-weighted unemployment-gap cell. M3 and M4 produce sizable holdout losses in both unemployment-gap cells. No extension therefore improves prediction consistently across cycle definitions, activities, and scoring rules. Core-CPI Pareto values are below 0.7, so its LOO calculation is more stable than the PPI comparison, but it still favors parsimony.

Core CPI model comparison

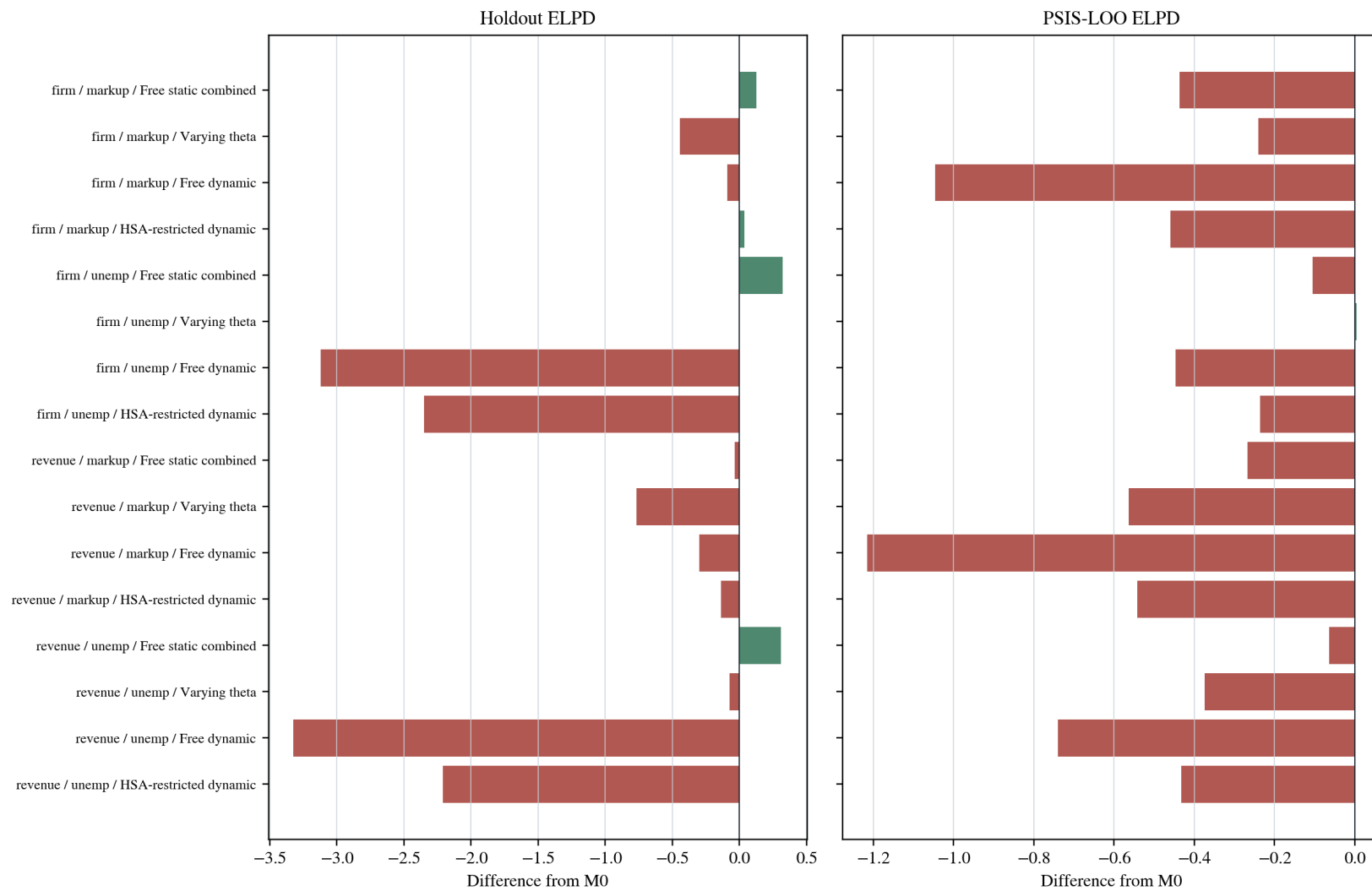


Figure 17: Core CPI predictive comparison relative to M0.

Table 25: Core CPI IID versus persistent AR(1), firm-weighted unemployment-gap cell

Model	Parameter	IID	Persistent AR(1)
Constant theta	theta.CIQ	-0.016 [-0.111, 0.078]	-0.021 [-0.106, 0.064]
Free static combined	delta	-0.016 [-0.067, 0.037]	-0.009 [-0.063, 0.041]
Free static combined	theta.CIQ	-0.029 [-0.133, 0.072]	-0.027 [-0.126, 0.067]
Varying theta	theta.0	-0.026 [-0.121, 0.069]	-0.029 [-0.116, 0.060]
Varying theta	gamma	0.030 [-0.025, 0.087]	0.027 [-0.023, 0.079]
Free dynamic	delta.1	-0.023 [-0.081, 0.034]	-0.018 [-0.075, 0.039]
Free dynamic	delta.2	-0.024 [-0.097, 0.048]	-0.022 [-0.090, 0.050]
Free dynamic	theta.0	-0.028 [-0.138, 0.088]	-0.028 [-0.134, 0.077]
Free dynamic	gamma	0.018 [-0.048, 0.083]	0.016 [-0.046, 0.082]
HSA-restricted dynamic	theta.0	-0.016 [-0.111, 0.044]	-0.014 [-0.103, 0.046]
HSA-restricted dynamic	gamma	0.009 [-0.057, 0.071]	0.009 [-0.051, 0.064]
HSA-restricted dynamic	lambda	0.564 [-8.287, 9.120]	0.524 [-7.949, 9.008]

AR(1) errors do not restore the HSA signs. The complete observed Core-CPI run has maximum R-hat 1.0036 and minimum bulk ESS 1,220.2.

Table 26: Core CPI observed IID convergence

Model	Cell	Max R-hat	Min bulk ESS
Constant theta	Firm / unemployment	1.0013	2506.3
Constant theta	Revenue / unemployment	1.0013	2565.3
Constant theta	Firm / inverse markup	1.0002	2496.8
Constant theta	Revenue / inverse markup	1.0008	2720.7
Free static combined	Firm / unemployment	1.0015	2272.7
Free static combined	Revenue / unemployment	1.0009	2722.5
Free static combined	Firm / inverse markup	1.0027	2534.3
Free static combined	Revenue / inverse markup	1.0011	2486.9
Varying theta	Firm / unemployment	1.0006	6664.0
Varying theta	Revenue / unemployment	1.0007	6284.3
Varying theta	Firm / inverse markup	1.0005	6442.0
Varying theta	Revenue / inverse markup	1.0006	6538.9
Free dynamic	Firm / unemployment	1.0008	6788.1
Free dynamic	Revenue / unemployment	1.0010	6648.0
Free dynamic	Firm / inverse markup	1.0012	6200.0
Free dynamic	Revenue / inverse markup	1.0012	6388.3
HSA-restricted dynamic	Firm / unemployment	1.0016	1413.2
HSA-restricted dynamic	Revenue / unemployment	1.0021	1639.9
HSA-restricted dynamic	Firm / inverse markup	1.0003	3825.2
HSA-restricted dynamic	Revenue / inverse markup	1.0004	3951.1

The computational conclusion is unambiguous: the Core-CPI sign differences and wide HSA intervals are posterior features, not nonconvergence artifacts.

28 Prespecified Oil-Control Extension

28.1 Why this is an admissible robustness exercise

QoQ PPI contains a large input-cost component, especially around the 2008–09 oil-price collapse and rebound. If that component is omitted, it can inflate the NKPC residual variance and reduce precision. It can also bias a competition coefficient if oil-price changes happen to covary with the competition regressors. The extension was therefore fixed before it was run: one oil series, one transformation, and exactly the current and first lag. No result-based lag selection or sign restriction was allowed.

Let R_t^o denote the repository's FRED-derived real WTI-to-CPI index and define

$$q_t^o = 400 (\log R_t^o - \log R_{t-1}^o) . \quad (38)$$

The normalization of the level cancels in the log difference. Every M0–M4 equation is augmented in exactly the same way:

$$\begin{aligned} \pi_t^q = & a + \alpha_b \pi_{t-1}^q + \alpha_f E_t \pi_{t+1}^q + \beta_{o,0} q_t^o + \beta_{o,1} q_{t-1}^o \\ & + \kappa_t x_t - \theta_t \hat{n}_t + \varepsilon_t . \end{aligned} \quad (39)$$

The definitions of κ_t and θ_t remain those of M0–M4 in Section 7. Thus oil is a conditional input-cost control, not an additional competition channel, and the competition-state posterior remains cut from inflation.

The oil priors are zero-mean Gaussian. For $j \in \{0, 1\}$,

$$\beta_{o,j} \sim N \left(0, \left[\frac{s_\pi}{s_{q_j^o}} \right]^2 \right) . \quad (40)$$

This gives the effect of a one-SD oil movement a prior SD of one outcome SD. The prior is weak relative to the observed PPI pass-through and is applied by the same scale rule to Core CPI.

Table 27: Oil-control data summary

Series	Source	Transformation	Start	End	T	Mean	SD
Oil control	FRED-derived real WTI/CPI index	$400\Delta \log R_t^o$	1993Q2	2013Q4	83	5.295	54.921
Oil control, lag 1	Same series	q_{t-1}^o	1993Q2	2013Q4	83	5.523	54.791

Two cautions remain. First, the input is a real WTI/CPI index, so the CPI denominator makes it a relative input-price control rather than a pure nominal oil shock. Second, current oil prices are not instrumented and may respond to aggregate conditions. The coefficient is therefore conditional pass-through, not a causal oil-shock estimate.

Prespecified oil control and the two inflation outcomes

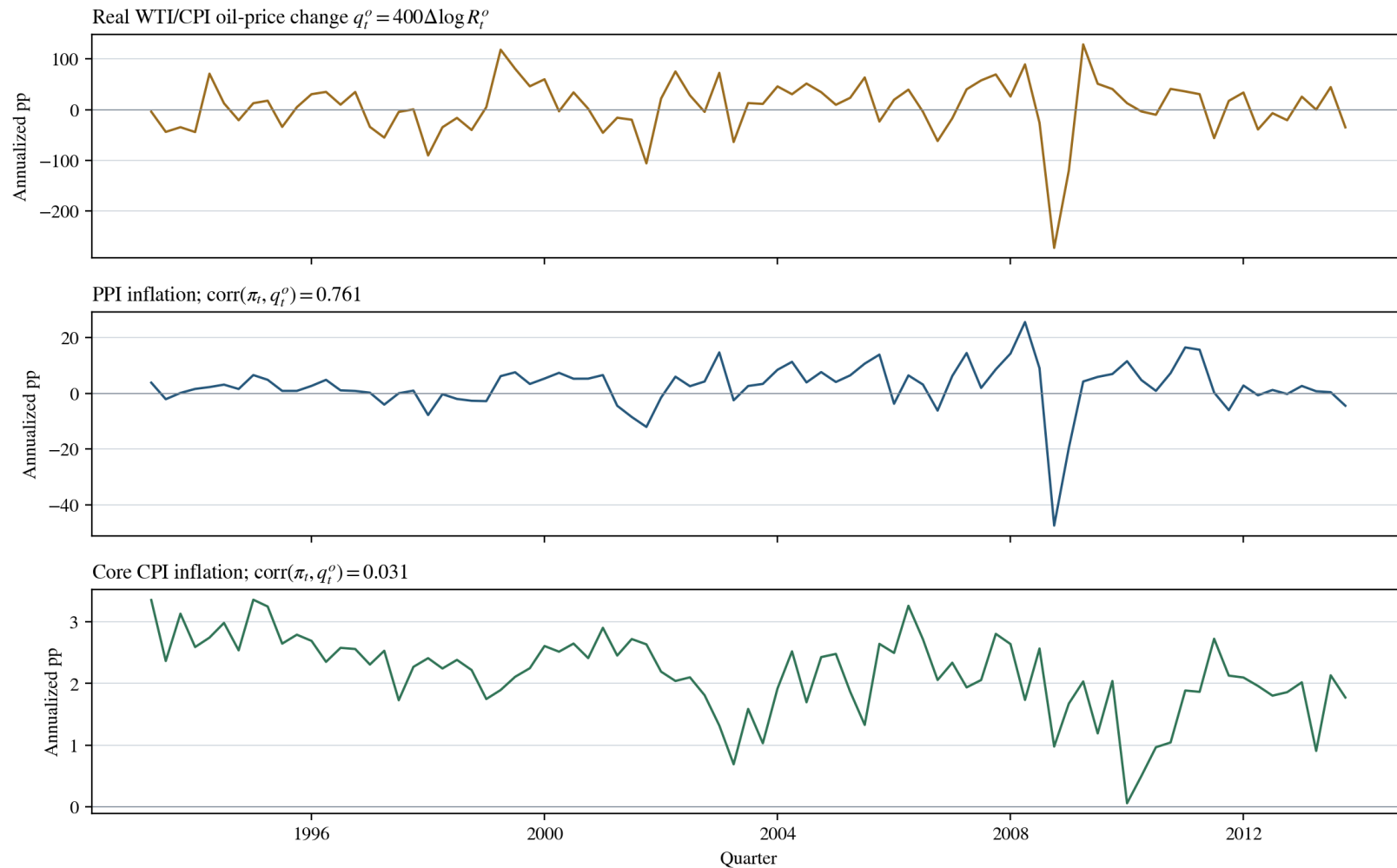


Figure 18: Prespecified real-oil-price control and the matched inflation outcomes.

The sample correlation of current oil-price change with PPI inflation is 0.761, compared with 0.031 for Core CPI. More importantly for omitted-variable bias, its correlations with the firm-weighted posterior-mean Capital-IQ cycle, the Gustavo slow state, and the negative unemployment gap are only -0.040 , -0.059 , and 0.019 . The oil extension therefore mostly removes PPI residual variation rather than projecting out the competition regressors.

29 Oil-Control Coefficients and Competition-Channel Stability

Oil-control prior versus posterior in M0

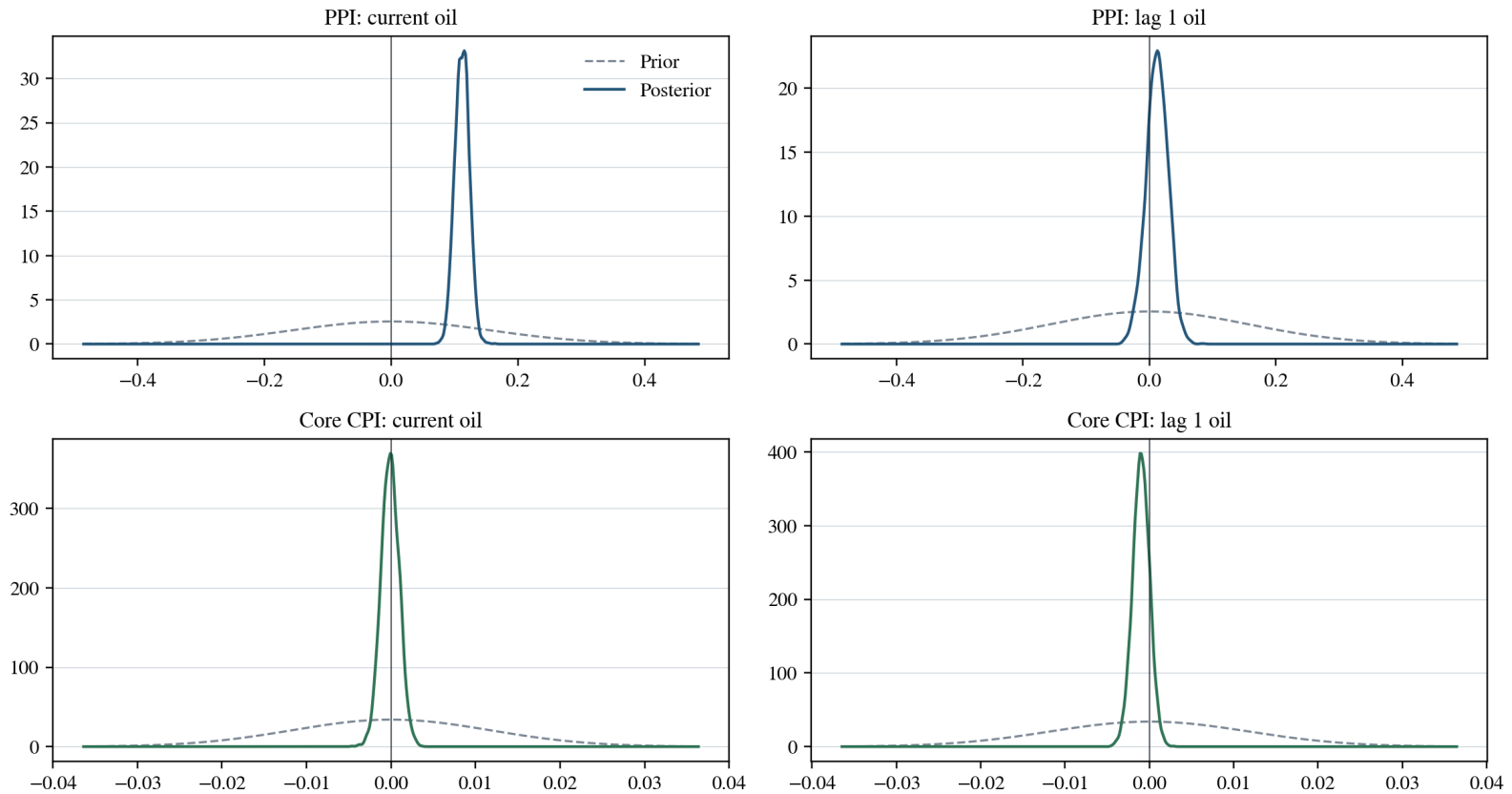


Figure 19: Prior versus posterior for current and lagged oil controls in M0.

For PPI the current coefficient is exceptionally well learned:

$$\beta_{o,0} = 0.112[0.091, 0.133], \quad \beta_{o,1} = 0.012[-0.023, 0.045]$$

in the firm-weighted unemployment-gap M0 cell. The lag-one coefficient is directionally positive but not separately sign-identified. For Core CPI both coefficients are economically near zero. This outcome difference is visible in the posterior densities and is not imposed by their priors.

Table 28: Oil-control coefficients and competition targets; firm-weighted cells

Target	PPI / unemployment	PPI / inverse markup	Core CPI / unemployment	Core CPI / inverse markup
M0 $\beta_{o,0}$	0.112 [0.091, 0.133]	0.111 [0.089, 0.133]	-0.000 [-0.002, 0.002]	-0.000 [-0.002, 0.002]
M0 $\beta_{o,1}$	0.012 [-0.023, 0.045]	0.012 [-0.022, 0.045]	-0.001 [-0.003, 0.001]	-0.001 [-0.003, 0.001]
M0 θ	0.691 [-0.391, 1.813]	0.723 [-0.310, 1.782]	-0.015 [-0.115, 0.078]	-0.022 [-0.123, 0.070]
M1 δ	0.072 [-0.538, 0.710]	0.042 [-2.384, 2.421]	-0.013 [-0.068, 0.039]	0.029 [-0.213, 0.266]
M1 θ	0.775 [-0.443, 2.016]	0.732 [-0.374, 1.840]	-0.025 [-0.129, 0.078]	-0.026 [-0.124, 0.072]
M2 θ_0	0.731 [-0.374, 1.853]	0.772 [-0.375, 1.933]	-0.024 [-0.121, 0.072]	-0.033 [-0.130, 0.063]
M2 γ	-0.098 [-0.714, 0.518]	-0.109 [-0.710, 0.506]	0.029 [-0.027, 0.086]	0.022 [-0.033, 0.077]
M4 λ	0.425 [-4.024, 5.513]	0.260 [-9.893, 10.725]	0.611 [-7.466, 8.870]	0.944 [-9.852, 12.058]

The central before/after comparison is:

Table 29: Primary unemployment-gap coefficients before and after oil control

Target	PPI: no oil	PPI: oil	Core: no oil	Core: oil
M0 θ	0.674 [-0.845, 2.200]	0.691 [-0.391, 1.813]	-0.016 [-0.111, 0.078]	-0.015 [-0.115, 0.078]
M1 δ	0.173 [-0.517, 0.861]	0.072 [-0.538, 0.710]	-0.016 [-0.067, 0.037]	-0.013 [-0.068, 0.039]
M1 θ	0.784 [-0.832, 2.388]	0.775 [-0.443, 2.016]	-0.029 [-0.133, 0.072]	-0.025 [-0.129, 0.078]
M2 γ	-0.165 [-1.030, 0.707]	-0.098 [-0.714, 0.518]	0.030 [-0.025, 0.087]	0.029 [-0.027, 0.086]
M4 λ	0.402 [-5.288, 5.952]	0.425 [-4.024, 5.513]	0.564 [-8.287, 9.120]	0.611 [-7.466, 8.870]

For PPI M0, the direct-loading mean is nearly unchanged, moving from 0.674 to 0.691. Its 95% interval narrows from $[-0.845, 2.200]$ to $[-0.391, 1.813]$, its posterior/prior SD ratio falls from 0.549 to 0.403, and $P(\theta > 0)$ rises from 0.816 to 0.890. The correct reading is that oil removes residual noise; it does not explain away the direct-loading mean. Because the 95% interval still includes zero, this is stronger directional evidence rather than strong sign identification.

M1 answers the next nesting question. With oil and a freely estimated slope channel,

$$\delta = 0.072[-0.538, 0.710], \quad \theta = 0.775[-0.443, 2.016], \quad P(\theta > 0) = 0.896.$$

The direct loading survives almost unchanged, while δ falls toward zero relative to its no-oil mean of 0.173. Thus the PPI data continue to prefer a direct direction over a slow-state slope interaction.

Competition coefficients before and after oil control

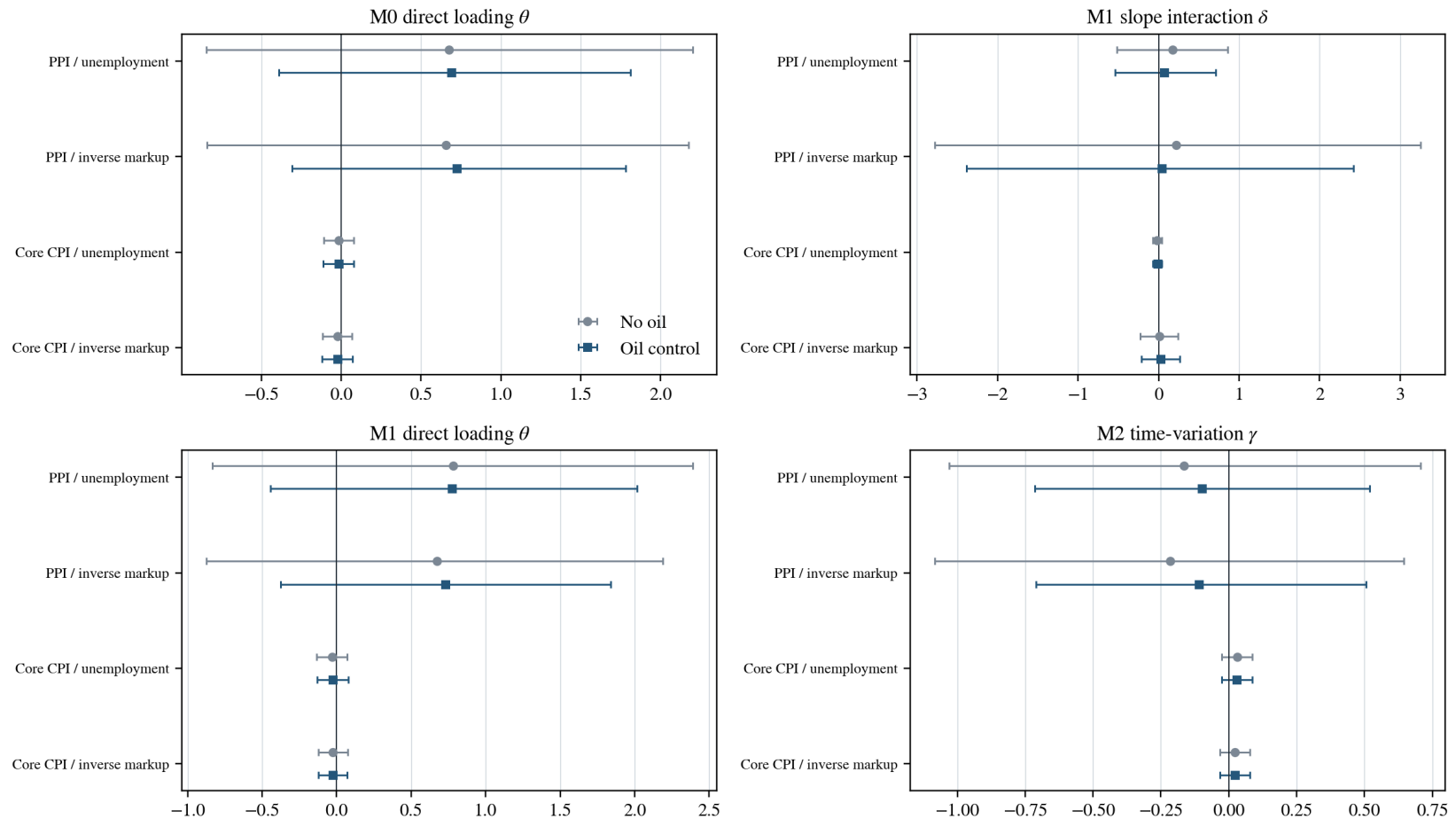


Figure 20: Competition coefficients before and after the fixed oil-control extension.

M2 does not reveal time variation after oil is included:

$$\theta_0 = 0.731[-0.374, 1.853], \quad \gamma = -0.098[-0.714, 0.518].$$

M3 remains weak, and M4 still gives $\lambda = 0.425[-4.024, 5.513]$ with zero-crossing derived slope coefficients. The oil control improves the inflation residual equation but cannot solve the multiplicative identification of $\lambda\theta_0$ and $\lambda\gamma$.

Core CPI is a useful placebo-like comparison. Its M0 loading changes only from $-0.016[-0.111, 0.078]$ to $-0.015[-0.115, 0.078]$. Its M2 γ remains directionally positive but zero-crossing. The same oil terms therefore do not manufacture a common HSA sign across price indices.

30 Oil-Control Prediction, Recovery, and Convergence

Oil-control predictive differences from the no-oil specification

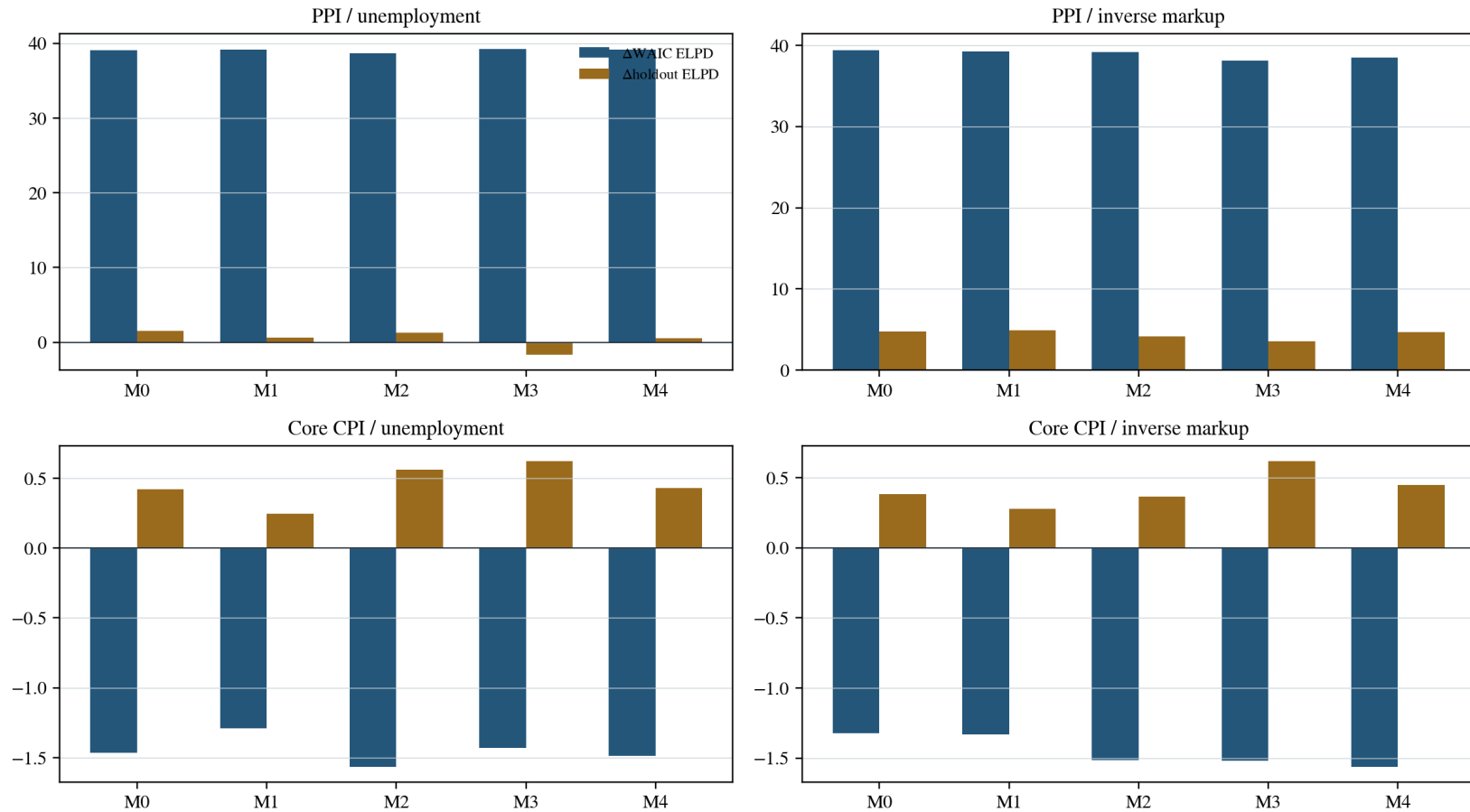


Figure 21: Oil-control predictive differences relative to each otherwise identical no-oil model.

PPI WAIC improves by roughly 38–39 ELPD units across M0–M4. Its PSIS-LOO comparisons point in the same direction, but maximum Pareto k remains above one, so the LOO magnitudes are not used as decisive evidence. Holdout behavior is more informative and less uniform. With inverse markup, oil raises holdout ELPD by 3.53–4.91 and lowers RMSE by 0.58–1.01. With the unemployment gap, M0 raises holdout ELPD by 1.55 but raises RMSE by 0.21, while

M3 deteriorates materially. Oil is clearly useful for fitting PPI, but its out-of-sample benefit depends on the activity proxy and model complexity.

For Core CPI, WAIC falls by 1.3–1.6. Holdout ELPD rises by only 0.25–0.62 and RMSE changes by less than 0.015. These small, mixed changes do not justify describing oil as an important Core-CPI control in this sample.

Table 30: Oil versus no-oil predictive comparison; firm-weighted cells

Cell	Model	Δ WAIC	Δ holdout	Δ RMSE	max k
PPI / unemployment	Constant theta	39.077	1.555	0.213	1.20
PPI / unemployment	Free static combined	39.169	0.620	0.747	1.21
PPI / unemployment	Varying theta	38.648	1.254	0.334	1.28
PPI / unemployment	Free dynamic	39.213	-1.692	2.083	1.19
PPI / unemployment	HSA-restricted dynamic	39.201	0.565	1.076	1.25
PPI / inverse markup	Constant theta	39.420	4.802	-0.951	1.40
PPI / inverse markup	Free static combined	39.237	4.914	-1.008	1.28
PPI / inverse markup	Varying theta	39.178	4.152	-0.803	1.11
PPI / inverse markup	Free dynamic	38.088	3.534	-0.577	1.17
PPI / inverse markup	HSA-restricted dynamic	38.495	4.682	-0.814	1.50
Core CPI / unemployment	Constant theta	-1.466	0.423	-0.004	0.61
Core CPI / unemployment	Free static combined	-1.290	0.246	-0.000	0.64
Core CPI / unemployment	Varying theta	-1.566	0.561	-0.009	0.61
Core CPI / unemployment	Free dynamic	-1.429	0.623	-0.009	0.61
Core CPI / unemployment	HSA-restricted dynamic	-1.487	0.428	-0.003	0.62
Core CPI / inverse markup	Constant theta	-1.324	0.380	-0.003	0.53
Core CPI / inverse markup	Free static combined	-1.331	0.277	-0.002	0.56
Core CPI / inverse markup	Varying theta	-1.514	0.363	-0.007	0.49
Core CPI / inverse markup	Free dynamic	-1.520	0.618	-0.015	0.63
Core CPI / inverse markup	HSA-restricted dynamic	-1.562	0.449	-0.006	0.66

Recovery was rerun with the fitted oil contribution retained in the data-generating process. At the observed standardized static effects $(s_\delta, s_\theta) = (0.06, 0.11)$, propagated- state suggestive recovery is 0.05 for δ and 0.40 for PPI θ . Core-CPI rates are 0.05 and 0.55. At $s_\gamma = 0.10$, suggestive recovery is 0.60 for PPI and 0.35 for Core CPI.

Table 31: Oil-control propagated-state recovery at observed-scale effects

Outcome	Parameter	True effect	Suggestive	Strong	Coverage
PPI	δ	0.06	0.05	0.00	1.00
PPI	θ	0.11	0.40	0.00	1.00
PPI	γ	0.10	0.60	0.00	1.00
Core CPI	δ	0.06	0.05	0.00	1.00
Core CPI	θ	0.11	0.55	0.05	1.00
Core CPI	γ	0.10	0.35	0.05	1.00

These rates remain below a conventional power target. The realized PPI sample becomes more directional after oil control, but repeated-sample recovery still warns against treating that posterior as reliable structural sign identification.

Table 32: Oil-control convergence; firm-weighted unemployment-gap cells

Outcome	Model	Max R-hat	Min bulk ESS
PPI	Constant theta	1.0031	2523.0
PPI	Free static combined	1.0008	2625.8
PPI	Varying theta	1.0013	6481.0
PPI	Free dynamic	1.0005	5795.2
PPI	HSA-restricted dynamic	1.0008	3901.3
Core CPI	Constant theta	1.0021	2595.4
Core CPI	Free static combined	1.0020	2415.0
Core CPI	Varying theta	1.0004	6493.4
Core CPI	Free dynamic	1.0005	6488.4
Core CPI	HSA-restricted dynamic	1.0032	1755.0

All 90 observed fits pass: maximum R-hat is 1.0046 and minimum bulk ESS is 1,106.3. Across the 560 recovery fits, maximum R-hat is 1.0180 and minimum bulk ESS is 344.4. Persistent AR(1) errors leave the PPI M0 result essentially unchanged: $\theta = 0.690[-0.501, 1.916]$ and $\beta_{o,0} = 0.112[0.089, 0.134]$. The substantive limits are identification and predictive stability, not MCMC convergence.

31 Why the Dynamic HSA Exercise Does Not Succeed

The evidence points to several complementary explanations rather than one coding or sampling defect.

1. **Low signal-to-noise ratio.** QoQ PPI inflation has an SD of 8.59 annualized percentage points. Core CPI is smoother, but its estimated competition contributions are also small. In both outcomes the observed standardized direct and slope effects remain below the reliable recovery region.
2. **Short aggregate sample.** The NKPC has only 83 quarters. Dynamic models add several competition interactions while estimating inflation persistence, the forward term, the activity slope, and innovation variance. Aggregate time-series variation is limited.
3. **The forward coefficient is weakly learned.** α_f retains nearly its prior dispersion. Expectation and persistence uncertainty consumes variation that might otherwise identify small competition contributions.
4. **State uncertainty is real but not decisive.** Capital IQ measurement error is material. Yet oracle-state recovery remains weak at observed effects, showing that a perfectly known cycle would not by itself solve the problem.
5. **Empirical coordinate versus theoretical N .** Capital IQ inverse concentration is not literally the HSA active-firm stock. The negative firm-weighted slow loading and the near-zero revenue-weighted loading reveal an unstable level mapping. A weak structural result may reflect measurement mapping as well as lack of an economic mechanism.
6. **Multiplicative HSA weak identification.** The restricted likelihood depends on $\lambda\theta_0$ and $\lambda\gamma$. Near zero, λ is weakly identified and sign symmetries remain. A broad $N(0, 10^2)$ prior correctly exposes this rather than imposing a preferred sign or fixed value.
7. **The sign pattern is not stable across price indices.** PPI average direct loadings lean positive and PPI γ leans negative. Core CPI produces the reverse pattern. Because neither set excludes zero, pooling or selecting one outcome would not establish a common HSA sign.
8. **No catastrophic linear collinearity to fix.** Condition numbers are modest. Dropping one regressor solely to manufacture significance would not address the underlying signal limitation and would change the economic question.
9. **Activity and expectation proxies are imperfect.** GDP-deflator expectations are not PPI-specific, and headline-CPI expectations are not Core-CPI-specific. The aggregate inverse markup is a proxy for theoretical real marginal cost, while the unemployment gap is generated using an estimated natural rate. The present conditional regression does not instrument these variables.
10. **Dynamic models do not predict better consistently.** All twelve PPI dynamic holdout ELPD differences are negative. In Core CPI, tiny improvements for M1 or isolated M2 cells coexist with clear M3/M4 losses. The extra flexibility does not earn stable out-of-sample support.
11. **Oil control clarifies but does not complete identification.** The fixed oil terms explain a large share of QoQ PPI variation and narrow the PPI direct-loading posterior. They do not make its 95% interval exclude zero, raise observed-scale recovery to a reliable level, identify δ or γ , or resolve the multiplicative HSA restriction.

These points rule out a convenient rescue by longer MCMC alone. Longer chains reduce Monte Carlo error, but they do not increase economic variation or repair an uncertain measurement mapping.

32 Interpretation, Scope, and Next Empirical Steps

32.1 What may be carried forward

The QoQ switch is useful. It aligns horizons, removes mechanical YoY overlap, and produces direct coefficients that are visibly updated from symmetric zero-mean priors. The PPI direction is positive and the Core-CPI direction is negative; both survive adding a slow slope and the AR(1) error law. The robust statement is therefore about outcome-specific empirical updating, not a common structural sign.

32.2 What may not be claimed

The 95% intervals do not identify the theoretical sign for either outcome. γ , λ , and both HSA derived slope terms are not identified. There is no valid current claim that HSA dynamic beats CES in marginal likelihood, nor that λ is a demand elasticity. The time-varying median paths cannot be interpreted as identified historical evolution.

32.3 Admissible next steps

1. Move from aggregate $T = 83$ to industry-by-quarter variation while maintaining an inflation-cut competition measurement block. This is the most direct way to raise information without tuning priors to the desired sign.
2. Validate the mapping from Capital IQ inverse concentration to an active-firm stock using external firm-stock or entry/exit data. The mapping should be tested before assigning a structural interpretation to λ .
3. Use PPI-specific and Core-CPI-specific expectation measures if defensible historical series become available, and propagate expectation uncertainty rather than treating either proxy as exact.
4. Address activity endogeneity with an economically defensible instrument or explicit shock system. A more elaborate state law should not substitute for identification of x_t .
5. If a larger panel or longer comparable sample becomes available, rerun simulation recovery before the empirical HSA restriction. The minimum detectable θ , γ , and slope effects should remain predeclared.

The appropriate current specification for reporting remains constant theta separately by price index, described as outcome-specific directional evidence. For PPI, the prespecified current-plus-lagged oil control is the better residual specification; for Core CPI it is a reported robustness exercise rather than a necessary control. Free and restricted dynamic models remain diagnostics of what the aggregate data cannot yet distinguish.

33 Reproducibility and Audit Trail

The report uses only saved results from `tests/gustavo_state_capitaliq_cycle`. The competition measurement draws are paired and reused byte-for-byte across the inflation models. The staged and dynamic manifests record the same processed-data SHA-256 hash and the same two measurement-draw hashes.

The dynamic run contains 27 observed fits: 24 IID fits from two Capital IQ cycles, two activity measures, three dynamic models, and full/train samples, plus three full-sample AR(1) robustness fits. It also contains 300 varying-theta recovery fits. The preceding staged run contains 17 observed fits and 640 free-static recovery fits.

The Core-CPI extension contains 45 observed fits: 40 IID full/train fits from two cycles, two activities, and all five M0–M4 models, plus five full-sample AR(1) robustness fits. It also contains 480 free-static and 300 varying-theta recovery fits. The saved Core-CPI manifest records maximum observed R-hat 1.0036 and minimum bulk ESS 1,220.2. A separate 45-fit smoke run verified plumbing before the full run; its short-chain posterior is not used in this report.

The oil extension contains 90 observed fits: 80 IID full/train fits from two outcomes, two cycles, two activities, and all five M0–M4 models, plus ten full-sample AR(1) robustness fits. It also contains 560 recovery fits. The saved manifest records maximum observed R-hat 1.0046 and minimum bulk ESS 1,106.3; recovery maximum R-hat is 1.0180 and minimum bulk ESS 344.4. The state-draw hashes are unchanged from the no-oil runs.

Assets and tables are regenerated with:

```
PYTHONPATH=src:. python docs/report/qoq_gustavo_capitaliq/build_assets.py
```

The Core-CPI estimation is reproduced with:

```
PYTHONPATH=src:. python \
  tests/gustavo_state_capitaliq_cycle/run_core_cpi_validation.py \
  --mode full --workers 4
```

The oil-control estimation is reproduced with:

```
PYTHONPATH=src:. python \
  tests/gustavo_state_capitaliq_cycle/run_oil_control_validation.py \
  --mode full --workers 4
```

The LaTeX report is compiled from this directory with:

```
latexmk -pdf qoq_hsa_report.tex
```

The detailed numerical records remain in `RESULTS_STAGED.md`, `RESULTS_DYNAMIC.md`, `RESULTS_CORE_CPI.md`, and `RESULTS_OIL_CONTROL.md`; the frozen equations and interpretation rules remain in `SPECIFICATION.md`. Structural claims remain gated by identification and recovery rather than computational convergence alone.